



Sustainable Development Verified Impact Standard

IMPROVED COOKSTOVE GROUPED PROJECT IN PAPUA NEW GUINEA VALIDATION REPORT



Document Prepared by Earthood Services Private Limited

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Client	Tasman Environmental Markets Asia Pacific
Project Title	Improved Cookstove Grouped Project in Papua New Guinea
Project Location	Papua New Guinea

Summary:

Earthood Services Private Limited (hereafter referred to as Earthood) has been contracted by Tasman Environmental Markets Asia Pacific to conduct the validation of the grouped project – Improved Cookstove Grouped Project in Papua New Guinea (VCS ID 4416) regarding the relevant requirements of SD VISta Program Guide and Standard (Sustainable Development Verified Impact Standard Version 1.0/01/ & SD VISta Program Guide Version 1.0/2.1/).

The proposed grouped project contributes to the achievements of SDG target 1, 3, 4, 5, 7, 8, 13 and 15. The project entails distribution and installation of fuel efficient improved cookstoves (ICS) for cooking purposes in the households of the host country, Papua New Guinea, thereby reducing the consumption of non-renewable biomass (fuelwood) used in inefficient traditional stoves. The project is a voluntary initiative as the host country's laws do not mandate the distribution of improved cookstoves (ICS) to the local communities.

The objective of validation is to perform a complete and independent review of the proposed project activity against the applicable SD VISta requirements, including the SDGs claimed in the project, people and planet benefits impacted by the project and compliance with the relevant SD VISta and host party criteria. These are assessed to ensure that the project design, as documented, is sound and reasonable, and that it fits the established standards. Validation is a prerequisite for all SD VISta projects, and it is viewed as vital to give stakeholders assurance about the project's quality and expected emission reductions. The VVB's aim is to perform a complete, independent evaluation of the project activity's validation.

The scope of validation can be described as an objective and independent review of the SD VISta Project Description (PD)/05/. The SD VISta PD is compared to the relevant criteria and guidance documents provided by SD VISta, including the SD VISta Program Guide (v1.0, dated 22 January 2019)/2.1/, Sustainable Development Verified Impact Standard (v1.0, dated 22 January 2019)/01/ and Program Definitions (v1.0, dated 22 January 2019)/2.2/ to ensure that the project meets the applicability conditions as per these documents.

The validation process is divided into 4 stages:

1. Desk review of the project description documents.
2. A review of data and information
3. Cross-checks between information provided in the SD VISta PD/05/ and information from sources using all available resources without regard to the project proponent's information.
4. Follow-up actions, including:
 - ❖ Online interviews/7/11/ with relevant stakeholders in the host country during the VVB remote audit with individuals having expertise of project development.
 - ❖ Cross-checking the information provided by interviewed people with the information provided by the project proponent using all available means.
 - ❖ Reference to publicly available information about projects or technologies that are similar to the grouped project under validation and review, depending on the approved methodology being used to ensure that formulae are acceptable, and calculations are accurate.

❖ The final validation report and opinion, as well as the resolution of lingering difficulties.

A risk-based approach has been followed to perform this validation. During the validation process, a total of 8 findings were raised, which includes 03 corrective Action quests (CARs), 04 Clarification Requests (CLs) and 01 Forward Action Request (FAR). All of these were successfully closed.

It is found that the SD VISTa PD/05/, and accompanying documents provided are all in compliance with SD VISTa criteria. The validation was completed with a reasonable level of assurance and no uncertainties were found related to the project activity validation.

Earthood Services Private Limited (hereafter referred to as Earthood) certifies that the grouped project “Improved Cookstove Grouped Project in Papua New Guinea” meets the requirements of the SD VISTa Project Description template, v1.0/04/, the SD VISTa Program Guide and Standard (Sustainable Development Verified Impact Standard Version 1.0/01/ & SD VISTa Program Guide Version 1.0/2.1/). The grouped project will contribute to a total of 8 SDGs which are SDG target 1, 3, 4, 5, 7, 8, 13 and 15. Earthood issues a positive validation opinion, certifying that the project meets the applicable Sustainable Development Verified Impact Standard and thereby proposing that it be registered.

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1 VALIDATION PROCESS

1.1 Objective

Tasman Environmental Markets Asia Pacific (PP) has contracted Earthood Services Private Limited (VVB) to carry out the validation of the grouped project activity “Improved Cookstove Grouped Project in Papua New Guinea” (VCS ID:4416) located in Papua New Guinea with respect to the relevant requirements of SD VISta Program Guide (v1.0, dated 22 January 2019)/2.1/, Sustainable Development Verified Impact Standard (v1.0, dated 22 January 2019)/01/ and Program Definitions (v1.0, dated 22 January 2019)/2.2/.

The purpose of this validation is to have an independent third-party assessment of proposed project activity to determine its compliance with the qualification criteria set out in the Sustainable Development Verified Impact Standard and Program Guide to attain real, measurable, additional, and permanent SD VISta claims. The validation statement opinion is a written assurance that:

- The project complies with all the applicable SD VISta requirements and can claim for SDGs over the project’s crediting period.
- The validation followed the requirements of the current version of the SD VISta Program Guide (v1.0, dated 22 January 2019)/2.1/ and Sustainable Development Verified Impact Standard (v1.0, dated 22 January 2019)/01/ to ensure the quality and consistency of the validation work and the report.
- The project shall result in addressing SDG target 1, 3, 4, 5, 7, 8, 13 and 15 by distributing 100,000 (ICS) and emission reductions as declared by the organisation or GHG project’s GHG assertion.
- The data reported is accurate, complete, consistent, transparent, and free of material error or omission.

1.2 Scope and Criteria

The scope of the validation is the review of the sustainable development impacts generated by the project, their contribution to the UN Sustainable Development Goals (SDG) and the benefits for people and prosperity and the benefits for the planet that they imply. With this aim, the audit assessed the project design, its management, the conditions of the stakeholders and the natural and ecosystem services at the project start, as well as the plans design for the monitoring of the project’s impacts. The SD Vista PD is reviewed against the relevant criteria and decisions by the SD VISta Program. Earthood has employed a risk-based approach in the validation, focusing on the identification of significant risks and reliability of project monitoring and generation of emission reductions.

The validation is not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design.

The validation is carried out based on the following requirements, applicable for this SD VISta grouped project:

- SD VISta Program Guide (v1.0, dated 22/01/2019) /2.1/
- Sustainable Development Verified Impact Standard (v1.0, dated 22/01/2019) /01/
- SD VISta Program Definitions (v1.0, dated 22/01/2019) /2.2/
- Other relevant rules, including that grouped project activity is in accordance with rules and legislation of the host country (Papua New Guinea).

The validation process is divided into four stages:

I. Desk review of SD VISta Project Description (PD):

- Evaluation of data and information
- Cross-checks between information provided in the SD VISta PD/05/ and information obtained from other sources using all available resources without regard to the project proponent's information.

II. Remote Audit- and follow-up interviews with project stakeholders:

- Remote interviews with relevant stakeholders in the host country with expertise in project development. /7/11/
- Cross-checking the information provided by interviewed people with the information provided by the project proponent using all available means.

III. Reference to publicly available information about projects or technologies that are similar to the grouped project under validation and review.

IV. The final validation report and opinion, as well as the resolution of outstanding issues.

1.3 Level of Assurance

☒ Reasonable Level of Assurance

☐ Limited Level of Assurance

The threshold for quantitative materiality with respect to the aggregate of errors, omissions and misrepresentations, individually or in the aggregate, for the reported SD VISta claim was limited to five percent, as required by section 5.2.3 of the Sustainable Development Verified Impact Standard /01/.

The approach used by Earthood for validation of the crediting period is built on a thorough understanding of the risks associated with reporting data on addressing SDG targets. Earthood conducted the validation by reviewing substantiating evidence and other relevant information and explanations from sources to provide reasonable assurance that targeted SDGs are fairly reported.

In the draft validation report (prepared by assessment team), the information provided is reviewed by an independent technical review team (one or more members) to confirm if the internal procedures

established and implemented by Earthood were duly complied with and such opinion/conclusion is reached in an objective manner that complies with the applicable SD VISta requirements as appropriate. The technical review team is collectively required to possess technical expertise of all the technical area/sectoral scope the project activity relates to. All team members of the technical review team are independent of the validation team. The report approved by the Technical Manager is endorsed by the Managing Director, who is overall responsible for ensuring quality, before final release. Further details of applicable procedures and responsibilities concerning the Earthood Quality Management System (QMS) are available on its website (www.earthood.in).

1.4 Summary Description of the Project

Summary description of grouped project activity has been illustrated in section 2.1.1 of SD VISta PDD appropriately. As per the given details in PDD, information was assessed remotely and found to be consistent. The project activity “Improved Cookstove Grouped Project in Papua New Guinea” undertakes dissemination of fuel efficient improved cookstoves (ICS) to households in Papua New Guinea free-of-cost. The grouped project activity is a voluntary initiative as host country laws do not mandate the distribution of ICS to the households. This ICS technology is designed to reduce GHG emission by improving the thermal efficiency of the cookstove thereby reducing the fuel consumed by households for domestic purposes and alongside promoting public health to tackle several health challenges such as indoor air pollution caused due to using less efficient stoves.

The baseline and project scenarios are well described and consistent with the data that was verified during the VVB remote audit. The average annual and total GHG emission reduction from the group project activity is expected to be 242,804 and 2,428,045 tCO₂e, respectively, over the crediting period.

Along with reducing emissions the project also aims at a holistic development of the community and planet. It does so by contributing to UN SDG Goals 1, 3, 4, 5, 7, 8, 13, and 15 which include in their purview increased savings, education, job creation, women empowerment, access to clean energy, reduced consumption of non-renewable biomass (charcoal) and decreased rates of deforestation in Papua New Guinea.

The project caters to the following SD VISta sectoral scopes:

Sectoral Scope 1 – Agriculture Forestry and Other Land Use

Sectoral Scope 2 - Climate Change Adaptation

Sectoral Scope 3 - Education

Sectoral Scope 4 – Energy

Sectoral Scope 7 - Health

Sectoral Scope 10 – Livelihoods

Sectoral Scope 14 – Women's Empowerment

1.5 Audit Team Composition

The Audit Team has been designed with much care to fulfil the set competency criteria. The team collectively has knowledge of the rules and requirements of the registry (VCS), sectoral scopes relevant to the project (03- Energy Demand), domain knowledge (SDGs etc.) and of the social and cultural expertise require for the project location (Papua New Guinea).

Earthood confirms that the audit team has no conflict of interest and furthermore is fully independent from all other aspects of the project.

The Validation team consists of the following personnel:

ROLE	LAST NAME	FIRST NAME	COUNTRY
Team Leader	Varshney	Divij	India
Trainee Validator	Yadav	Vaishali	India
T.A Expert (3.1)	Varshney	Divij	India
Local Expert	Ogil	Jewel	Papua New Guinea
Technical Reviewer	Guleria	Shifali	India
T.A Expert to T.A(3.1)			

The UN SD Experts involved in the project are listed below.

ROLE	LAST NAME	FIRST NAME	UN SDG
External Expert SD VISta Scope 1- AFOLU	Srivastava	Parul	UN SDG 15
External Expert SD VISta Scope 2- Climate Change Adaptation	Varshney	Divij	UN SDG 13
External Expert SD VISta Scope 3- Education	Kumar	Avinash	UN SDG 4
External Expert SD VISta Scope 4- Energy	Singh	Ranjan	UN SDG 7
External Expert SD VISta Scope 7- Health	Kumar	Avinash	UN SDG 3
External Expert SD VISta Scope 10- Livelihoods	Kaushik	Sumit	UN SDG 1 UN SDG 8
External Expert SD VISta Scope 14-	Mahala	Deepika	UN SDG 5

Women Empowerment			
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The Competency Statement of each team member, explicated, can be found in Appendix I.

1.6 Method and Criteria

Earthood Services Private Limited has been contracted by the Tasman Environmental Markets Asia Pacific to conduct the validation of the project activity, with regards to the relevant requirements of Sustainable Development Verified Impact Standard, Version 1/01/. The project activity uses the VMR0006 version 1.2/08/, which is classified as sectoral scope 3 “Energy Demand”. The following were included in the validation:

- Contract with PP for the scope and appointment of validation team and technical review team.
- Completeness check of VCS PD.
- Project Description desk review and estimated GHG emission reduction calculation sheet.
- Remote inspection by validation team.
- Interview with project representatives and project owner.
- Reporting and closure of findings (CARs/CLs/FARs) and preparation of draft validation report.
- Independent technical review of the draft validation report and final/revised documentation (e.g., VCS PD, corresponding estimated ER calculations sheet and evidence).
- Issuance of the final validation report to the contracted PP.

Sampling plan for validation of the project:

The Sampling and surveys for CDM project activities and programmes of activities (Version 9.0)/12/ states under paragraph 28 that “When the project participants or the coordinating/managing entity have applied a sampling approach, the VVB may apply acceptance sampling as described in the steps indicated in paragraphs 29–38 as part of validation/verification activities”. The validation team conducted acceptance sampling in line with paragraph 30 and 31 of the sampling standard version 9.0/12/.

The validation team determined the sample size for acceptance sampling by evaluating the following, using its own professional judgement and guidance in the Standard ‘Sampling and surveys for CDM project activities and programme of activities, Version 9.0’:

- The proportion of discrepancies between the CME’s data and validation team’s (field or onsite inspection results) data that can be considered acceptable. This is referred to as the AQL (Acceptable Quality Level): 0.5% was considered in this assessment.
- The proportion of discrepancies between the CME’s data and validation team’s (field or onsite inspection results) data that would be considered unacceptable. This is the UQL (Unacceptable Quality Level): 20% was considered in this assessment.
- The producer risk and consumer risk of 10% each was considered.

The Verification team selected the sample size as 11 project ICS consumers with 3 additional consumers as backup for the purpose of remote inspection to check the acceptability of PP's sampling result or otherwise.

The verification team selected the random samples from PP's baseline survey and pilot project sampled records to check the acceptability (or otherwise) of the data for each such record with PP's baseline survey and pilot project sample records and determine if the PP's sampling results is acceptable or otherwise.

Sample Size:

AQL	UQL	Producer Risk	Consumer Risk	Sample Size; Min	Acceptance No.
0.5%	20%	10%	10%	11	0

The Validation team covered a total of 11 samples to confirm baseline scenario and verify the baseline technology used previously, the fuel used in baseline and project device and to check the implementation of the improved stoves. The assessment team has used remote audit interviews, surveys etc. to support the validation criteria to reach a reasonable level of assurance. Earthood can confirm that the same, and also confirmed the pilot project is implemented in line with the description in the SD VISTA PD /5/. Most of the households interviewed were found to have inefficient three stone stove as baseline stove using firewood which were replaced by the project stove in case of end-users. The ICSs disseminated had unique serial number and the end users confirmed that they have stopped using the three-stone fire stove and the cookstove was distributed free-of-cost to the households. A few households which received the ICS in the pilot-run admitted to own LPG and electric stoves as well, but their usage depended on the unreliable electricity and gas availability in the region. A finding CL#04 was raised in this regard. No other inconsistencies were observed. The end user details were also confirmed and found to be the same as the information provided in the distribution database provided by the PP/30/.

1.7 Document Review

According to the Sustainable Development Verified Impact Standard requirements/01/, the SD VISTA project description/05/, and supporting documents related to the project design and baseline were reviewed. The desk review included:

- Evaluation of the data and information given to ensure that it is complete and consistent with the Sustainable Development Verified Impact Standard/01/.
- A review of the project description/05/, with special emphasis to the U.N. Sustainable Development Goals.
- Review of the monitoring plan, stakeholder identification and engagement in the project.

In addition, the validation team employed third-party material such as host-party legislation, technical reports pertaining to the project design or basic conditions, and technical data available in the public domain. The validation team cross checked the information, statements and claims in the PD through various evidence provided by PP or from other sources. A complete list of documents/ evidence reviewed is included as Appendix III.

Earthood conducted a desk review as under.

- a. A review of the data and information presented to verify their completeness.
- b. A review of the monitoring plan, the monitoring methodology including applicable tool(s) and, where applicable, the applied standardized baseline, paying particular attention to the frequency of measurements, the quality of project technology including efficiency of the project stove, and the quality assurance and quality control procedures
- c. An evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions

1.8 Interviews

Though SD VISta Program Guide (v1.0, dated 22 January 2019)/2.1/, para 3.5.6 says, “Assessments shall include a visit to the project site. The purpose of the site visit is to confirm the validity of the project description or monitoring report and to ensure that a project meets the rules and criteria of the SD VISta Program”, due to security concerns, the VVB has conducted a remote audit on 22/09/2023 to check the baseline scenario and implementation plan of the project activity. A third-party security risk assessment by SecRisk has been conducted and has been shared by PP/33/, through which The Letter of Exemption for the on-site visit has been received from Verra /40/. The remote visit was conducted as a part of validating the project and to gain additional information regarding the compliance of the project with the SD VISta requirements. It was conducted to check the baseline technology used previously, the fuel used in baseline and project device and to check the implementation of the improved stoves in the pilot project, along with the interviews of the LSC personnels and project proponent representatives to review the sustainable development benefits that the project intends to provide. PP has conducted the baseline survey in 220 households across Western Highlands Province (WHP) and Southern Highlands Province (SHP) in Papua New Guinea (PNG) and has also installed a pilot cookstove project in these households for the pilot study of the cookstove. During the Remote Audit, the end-users were interviewed regarding the baseline technology they used, the current fuel type, the project technology they acquired, to check the additionality asked them if they have received the cookstove free of cost and so on. The following points were discussed as a part of the interviews:

- a) Introduction, scope and objective of work, roles and responsibilities, resources required, and timetable of the Remote audit including venue for closing meeting and any concerns from PP.
- b) All the details about the project activity mentioned in the PD were cross-checked by discussion with the PP.
- c) Conducting the remote audit interviewing the end-users
- d) And final submission of the audit findings to the client and agreement on the issues raised.

The table below describes the remote interview process and further identities personnel, including their roles, who were interviewed and/or provided information additional to that provided in the project description and any supporting document.

S.No.	Name	Organization	Date	Topic	Team
1.	Gabriel Machovsky Capuska	Tasman Environmental Markets Asia Pacific Pte. Ltd	22/09/2023	Project design, Project Implementation status, Project start date and Project Location, Baseline Scenario, Baseline Identification and Additionality, Qualification and Training, Monitoring and reporting documentation, Quality Assurance, Management and operating systems, local stakeholders meeting process, Compliance with relevant laws, Roles and responsibility	Divij Varshney, Vaishali Yadav, Jewel Ogil
2.	Cherie Sim	Tasman Environmental Markets Asia Pacific Pte. Ltd	22/09/2023		
3.	Michael Kisombo	Tasman Environmental Markets Asia Pacific Pte. Ltd	22/09/2023		

Interview with the end users

S.No.	Name	Organization/ Affiliation	Date	Topic	Team member
1.	Donald Yaget	End User- Western Highlands Province (WHP)	22/09/2023	Project stove Installation, Baseline stove and baseline fuel, Fuel used in project stove, Training and ongoing support by the PP for operation of the project stove, Project stove usage patterns, and Grievances.	Vaishali Yadav, Jewel Ogil
2.	Betty Kaika	End User- WHP	22/09/2023		Vaishali Yadav, Jewel Ogil
3.	Loise Kuli	End User- WHP	22/09/2023		Vaishali Yadav, Jewel Ogil
4.	Sarah Yak	End User- WHP	22/09/2023		Vaishali Yadav, Jewel Ogil

5.	Mekail	End User- WHP	22/09/2023		Vaishali Yadav, Jewel Ogil
6.	Jonathan James	End User- WHP	22/09/2023		Vaishali Yadav, Jewel Ogil
7.	Jerman Okipul	End User- Southern Highlands Province (SHP)	22/09/2023		Divij Varshney, Jewel Ogil
8.	Clancy Sendil	End User- SHP	22/09/2023		Divij Varshney, Jewel Ogil
9.	Yara Ungugomo	End User- SHP	22/09/2023		Divij Varshney, Jewel Ogil
10.	Ps Nelson Larsawan	End User- SHP	22/09/2023		Divij Varshney, Jewel Ogil
11.	Anna Kenneth Thomas	End User- SHP	22/09/2023		Divij Varshney, Jewel Ogil
12.	Maria Joseph	End User- SHP	22/09/2023		Divij Varshney, Jewel Ogil

Additionally, to assess the capabilities of the ground staff to implement the project, 4 field assistants were remotely interviewed by the assessment team.

Interview with the Field Assistants

S.No.	Name	Organization/ Affiliation	Date	Topic	Team member
1.	Arnold Terry	Field Assistant	22/09/2023	Trainings provided, baseline survey, implementation survey, registration process.	Divij Varshney, Vaishali Yadav, Jewel Ogil
2.	Tony Yamba	Field Assistant	22/09/2023		Divij Varshney,

					Vaishali Yadav, Jewel Ogil
3.	Patrick Wapiyu	Field Assistant	22/09/2023		Divij Varshney, Vaishali Yadav, Jewel Ogil
4.	Michael Kisombo	Field Assistant	22/09/2023		Divij Varshney, Vaishali Yadav, Jewel Ogil

Interview with the Local Stakeholders Consultation (LSC) Meeting participants

S.No.	Name	Organization/ Affiliation	Date	Topic	Team member
1.	Ps Enoc Wan	LSC participant	22/09/2023	Benefits of the project activity, Feedback about the project, Grievance register.	Vaishali Yadav, Jewel Ogil
2.	George Herphit	LSC participant	22/09/2023		Vaishali Yadav, Jewel Ogil
3.	Janet Terry	LSC participant	22/09/2023		Vaishali Yadav, Jewel Ogil
4.	Magret Nip	LSC participant	22/09/2023		Divij Varshney, Jewel Ogil
5.	Peter Wapenda Wari	LSC participant	22/09/2023		Divij Varshney, Jewel Ogil
6.	Joel Raus	LSC participant	22/09/2023		Divij Varshney, Jewel Ogil

Through interviews and remote audit, the assessment team confirms the baseline scenario, project fuel type, additionality, local stakeholder consultation and its feedback. Thus, the assessment team achieved a reasonable level of assurance by conducting a remote site- visit of end-users.

1.9 Site Inspections

The VVB carried out remote audit on 22/09/2023 and inspected the project technical design and implementation as specified in the SD VISTA PD/5/ The remote assessment was carried out with the objective to:

- Determine whether the project activity meets all applicable SD Vista requirements, including those specified in the standard, relevant methodologies, tools, standardized baselines and guidelines.
- Assess the accuracy, conservativeness, relevance, completeness, consistency, and transparency of the information provided by project participants.
- Determine whether information provided by the project participants is reliable and credible.

The remote audit was conducted to validate the accuracy and completeness of the project description. The VVB team members who visited the site collectively had competence for the relevant technical area to which the project relates. The views obtained during site observation were considered while concluding the validation opinion.

The main topics covered during the remote audit interview are as follows:

- General aspects of the project and its implementation
- Installation and operation of ICS under the proposed grouped project
- Assessment of baseline scenario and project boundary
- Authentication of the SDG impacts and monitoring and controlling instruments and operational arrangements.
- Analysis of the monitoring provisions by checking the monitoring arrangement.

The Sampling plan adopted by the VVB for carrying out validation assessment has been discussed prior in section 1.6. The list of end users interviewed by the assessment team has been listed in section 1.8 of this report.

1.10 Public Comments

This project was open for public comment from 14/07/2023 to 13/08/2023. No comments were received by the project during the public comment period as confirmed from the project registry webpage /13/.

1.11 Resolution of Findings

This section summarizes the results of the project's validation process. The results of the document review, remote audit evaluations, and interviews are presented in this section. CARs, CLs, and FARs are used to correct material inconsistencies discovered during validation.

Corrective Action Request (CAR)

Participants in the project have committed mistakes that will affect the project's capacity to deliver meaningful, measurable further emission reductions.

- i. Specific methodological standards that are applicable have not been met, or
- ii. There's a chance that emission reductions won't be able to be measured or tracked.

Clarification Request (CL)

It's used when more information is needed to properly understand an issue or when the data isn't clear enough to determine whether a condition has been satisfied.

Forward Action Request (FAR)

- i. For the next verification period, the actual project monitoring and reporting processes demand attention and/or correction, or
- ii. It is recommended that the MP be adjusted.

CARs and CLs will be resolved or closed if the PP changes the project description, corrects the PD, or provides sufficient further explanations or proof to address the concerns. If this is not done, the project activity will not be recommended for VCS registry registration.

In summary, the type and total number of findings that were raised are indicated below.

Type of Finding	CAR	CL	FAR
Total Findings	03	04	01

All the findings raised during this validation of the project are included under Appendix IV of this report.

1.12 Forward Action Requests

01 Forward Action Request (FAR) was raised during this assessment. The Group project activity is undergoing Validation. During the validation process, an exemption has been taken by the project proponent to conduct the remote audit of the project activity due to the significant security risk in the vicinity of project area, which has been approved by the VERRA on conditional basis/40/. A FAR has been raised by the VVB to conduct the on-site visit of the project activity during the first verification period. The details of the same can be found in Appendix IV of this report.

2 VALIDATION FINDINGS

2.1 Summary of SDG Contributions

The grouped project “Improved Cookstove Grouped Project in Papua New Guinea” is aiming to reduce GHG emission by replacing traditional cookstoves with highly thermal efficient Improved Cookstoves, G3 Burn rocket stove. The proposed project will contribute to 8 SDGs as U.N. sustainable Development Goals 1, 3, 4, 5, 7, 8, 13 and 15.

The following table summarizes the project’s direct contribution to the SDG through the implementation of the project activities.

S No.	SDG Target	SDG Indicator	Estimated Project Contribution by the End of Project Lifetime and Net Impact on SDG Indicator	Assessment opinion
1.	Project specific indicator Goal 1	Goal 1: Amount of money saved by families that buy firewood for improved cookstoves distributed under the project activity	Reducing the economic burden of 100,000 Households / families that buy firewood for the ICS distributed under the project activity. Net Impact: Implement activities to increase	The validation team reviewed the project and found that 100,000 energy efficient ICS will be distributed by the PP to end users according to the project design. It will lead to a reduction in the amount of money spent in buying the firewood and increase in the total savings of the households. Clean cooking technology is the basic service provided by the project.
2.	3.9	Goal 3: Reduction of fine particulate matter (PM2.5) emissions within households to reduce air pollution within households in the project area.	The project activity plans to replace 100,000 traditional cookstoves with energy efficient project cookstoves will contribute to improved health and well-being brought about by reduced levels of fine particulate matter (PM2.5) emissions within households below the baseline emission level of 943.3 mg/MJ. Net Impact: Implement activities to increase	The assessment team confirms that the information provided in this table is complete with respect to SD VISTA requirement. The amount of PM 2.5 emission for the baseline stove has been taken from the clean cooking catalogue/56/, and hence, found to be appropriate. The estimated amount of PM2.5 emissions that were reduced as a result of project stove usage has not been provided here by PP. It was validated from remote audit interviews /11/ that smoke has reduced indoors with the usage of ICS.

3.	4.3	Goal 4.1: Number of youth and adults in formal and non-formal education and training in the previous 12 months, by sex	An estimated 130 local staff and personnel will be trained, and training courses will be conducted annually for all the staff for latest knowledge transfer Net Impact: Increase	It is validated through Training manual and training records/35, 32/ that PP is providing training to the local individuals from time to time regarding the stoves. The PP also plans to provide training to the end users through the enumerators about health, nutrition, environment etc. from time to time. It is validated through the remote interviews and desk review by the assessment team that due to use of ICS the children get more time for their education and the females (primary cooks of family) are now able to spend more time with family and income – generation activities. Thus, the project will achieve SDG goal 4. It was further clarified by the PP that the trainings are not biased towards one gender over another; instead, emphasis is placed on giving opportunities and trainings to women. Additionally, there are many women who are working and have received professional training within the workforce (enumerators and other field employees). Additionally, a significant number of women will receive the intended training throughout the project's lifespan.
4.	4.4	Goal 4.2: Number of youth and adults with technical and vocational skills, by type of skill within the project area	The PA aims to enhance the number of skilled youth available in the project locality, thus increasing their employability and income generation potential. For monitoring the PA, the PP plans to train around 130 local people and continue the knowledge transfer. Net Impact: Increase	It is validated from literature review of TEM's Employee Handbook/26/ that the organization encourages diversity. It was further corroborated through the remote audit interviews /11/ that people are being skilled. PP has shared the MoU signed with the Western Pacific University (WPU)/55/, which clearly states that Vocational and technical trainings and

				employment will be provided to the local university students and appointed staff in the area of sustainability, climate change impacts and mitigation strategies and improved cookstoves.
5.	5.4	SDI 5.4.1: Proportion of time spent on unpaid domestic and care work, by sex, age, and location	The burden of collecting and/or purchasing fuel often falls on women and children. By reducing cooking time, the program reduces women's drudgery and provides them with more time to invest in other productive economic development activities. The PA aims to reduce the collection frequency to 2-3 times/week Net Impact: Implement activities to decrease	It is validated from remote audit interview /11/ and desk review that due to use of ICS, women and children's drudgery time will be reduced. This is also validated by assessing the metrics in baseline survey/16 and monitoring survey, and the SDVista PD/5.
6.	5.5	Goal 5: Number of women serving in managerial/ leadership /ownership role	This project activity encourages participation of women in leadership / managerial role. Net Impact: Implement activities to increase	It is validated from remote audit interviews that the project contributes to generation of employment in informal sector (total economy, agriculture, and non-agriculture) by contracting locals with a focus on hiring females. The TEM Employee Handbook/26/ has been assessed to ascertain the PP's thrust towards equitable employment for women and marginalized groups and has been found appropriate and satisfactory. #CL03 was raised and satisfactorily closed with regard to the concerned SDG.
7.	7.1	Goal 7: Number of people with primary reliance on clean fuels and technology	The project activity involves promotion and distribution of 100,000 improved cooking stoves (ICS) in the households of Papua New Guinea. Net Impact: Increase	The validation team reviewed the project and found that 100,000 energy efficient ICS will be distributed by the PP to end users according to the project design. Clean cooking technology is the basic service provided by the project.

8.	8.3	Goal 8: Number of people formally employed by sector and sex under the project activity.	The project activity aims to provide training and skill development programs for the usage of ICS, and its contribution in mitigating climate change and the carbon cycle, thus increasing youth's employability. Net Impact: Implement activities to Increase	This is validated via assessment of training records, and training curriculum provided by PP. It is validated through Training material /35/ and SDVista PD/05/ that PP provides training on annual frequency basis. It is further validated from remote audit interview/7/11/ that PP has conducted trainings as part of youth and women skill development initiative in its region of implementation.
9.	8.5	SDI 8.5.1: Average hourly earnings of employees, by sex, age, occupation, and persons with disabilities	The project activity aims to generate an estimate of 10 full-time and 120 part-time local employment. Individuals employed under the project activity will likely be community members who are currently unemployed or working in the informal sector with an average income estimated as K439/month (Bonilla, et al., 2020). The project activity will compensate these employees fairly and above this rate. Net Impact: Implement activities to Increase	This is validated through the training manuals and the estimate provided in the SDVista PD /05/. It is further confirmed by assessment team during the remote audit/7/. The increased income will be checked at the time of verification through the employment records and employee payslips.
10.	13.0	Tons of greenhouse gas emissions avoided	242,804 tCO_{2e} of greenhouse gas emissions will be avoided annually over a 10-year period Net Impact: Increase	During assessment it is confirmed by validation team that the value 242,804 tCO_{2e} is correctly estimated which is validated through the assessment of the Ex-ante ER sheet/6/
11.	15.2	SDI 15.2.1: Progress towards sustainable forest management through the increase of above-ground biomass in forests	The Project will reduce an estimated amount of 3.81 tons of Non-renewable biomass per stove annually, which will contribute towards reducing deforestation and increasing above-ground biomass Net Impact: Implement activities to increase	The assessment team confirms that the information provided in this table is complete with respect to SD VISTA requirement. During the remote audit interview, it is confirmed by validation team that there is significant reduction in firewood usage by the end users since the implementation of the project.

				The amount of reduction in woody biomass has been estimated based on the applied methodology in VCS PD and is found to be consistent. The value has been checked through the ex-ante ER sheet/6/ and is found to be appropriate. CL#04 was raised and resolved.
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2.2 Project Design

2.2.1 Project Objectives

The project “Improved Cookstove Grouped Project in Papua New Guinea” aims to reduce GHG emission by replacing traditional cookstoves with highly thermal efficient cookstove, G3 Burn rocket stove. The Cookstove project has contributed to 8 SDGs out of total 17 Sustainable Development Goals (SDG). The project activity is classified as U.N. sustainable Development Goals 1, 3, 4, 5, 7, 8, 13 and 15.

The Project has clearly stated its objectives in Section 2.1.1 of the SD VISta PD/05/

The project activity aims to reduce the financial burden on the families that will be covered under this PA, by replacing 100,000 traditional cookstoves with fuel efficient improved cookstoves (ICS). It will lead to lesser consumption of firewood. The project activity also focuses on providing more job opportunities irrespective of people’s sex, age, occupation, and disabilities. This includes jobs in varied fields, and local people from different backgrounds are expected to get managerial positions as well as groundwork. The project aims to provide skill development and training to the local population to help them hone their talents and become ‘employable’. The project activity aims to recruit women in the workforce and envisages that 50% of its employees will be female.

Time to prepare food will decrease by using ICS because they are more efficient as compared to traditional cookstoves. The time saved will allow the female population, who are primarily involved in cooking, to invest time in other pursuits like education.

Ensuring the widespread availability of ICS (Improved Cookstoves) guarantees that the public can access clean energy sources. Additionally, enhancing the efficiency of biomass fuel usage reduces fuel consumption, thereby decreasing household spending on fuel wood and poverty. This efficiency also leads to lower greenhouse gas emissions, a reduction in biomass fuel usage, and less deforestation for fuelwood access.

The ICS’s efficiency has been validated from the technical specifications provided and the WBT test results/21/. Training to the local people is confirmed through training manual/35 and training records/32. Reduced inequality in the area is confirmed through equal work opportunity policies and non-discrimination policies of the PP/10, 14, 26/.

Project is expected to reduce GHG emission by 242,804 tCO₂ eq per year by distributing the G3 Burn rocket stove.

2.2.2 Project Activities

In the SD VISta PD section 2.1.2 Description of the project activity explained the project activity in detail. During assessment following points are concluded by the validation team which are mentioned below:

The project activity is a grouped project. The ICS will continue to use non-renewable biomass for cooking, but the ICS will use less wood fuel to meet thermal needs because it has higher thermal efficiency, which will result in a reduction of GHG emissions compared to the baseline scenario. It is replacing conventional cooking solutions with fuel-efficient improved cookstoves, which falls under the category of efficiency improvements in thermal applications.

The G3 Burn rocket stove is a domestic stove with thermal efficiency of 50.8% (with pot skirt). It has dimensions 32.1 * 28.2 H cms. It utilizes fuelwood and has a life span of 10 years. This is verified by the WBT test report of CREEC/21/ for G3 Burn rocket stove. It is estimated that 100,000 ICS will be distributed in Papua New Guinea during the project lifetime/05/.

The primary thrust of the project is distribution and installation of ICS. For the same, field staff have been provided with education and training so they may implement, monitor, and evaluate the project.

The project also provides training to locals to recruit them as staff, thus providing employment, and to also develop their employable skill which will allow them to find work, post culmination of project activity. The Project activity also results in uplifting women, by providing them employment opportunity and leadership roles. Additionally, women and girls being primarily responsible for collection of fuel and cooking, now can save time which can be used in other pursuits. The reduction in burn risk, indoor air pollution and long hours also ensures they have healthier lifestyles.

2.2.3 Implementation Schedule

Implementation schedule is discussed in section 2.1.3 of PD. The first important date is January 2024 which is when Tasman Environmental Markets Asia Pacific as an organization aims to start distributing ICS in the project area. The start date of the grouped project activity, as identified by the PP is 01/01/2024, which is also the date of installation of first ICS under the Grouped project activity. This is validated based on the planned date of distribution of first ICS under this grouped project activity/05/. The crediting period is fixed for ten years, beginning on 01/01/2024 to 31/12/2033.

The stakeholder meeting was held from 13/03/2023 to 20/03/2023. The start date is assessed from the stakeholder consultation reports, invitations and records/15/.

The baseline survey was conducted by PP in October-November 2022 which is verified from the baseline survey records /16/.

In July 2023, the project was listed under the VCS and SD VISta program which is verified from the project webpage on verra website -<https://registry.verra.org/app/projectDetail/VCS/4416>. /13/

2.2.4 Project Proponent and Other Entities Involved in the Project

Tasman Environmental Markets Asia Pacific is the project proponent. The parent entity 'Tasman Environmental Markets Pty Ltd' is financially supporting the PP in the development, management and ongoing monitoring processes for the project activity. The assessors have confirmed the same with the TEM letter of support signed.pdf /37/. Simi Oa is providing consultancy services to the PP, and it is the implementation partner in this project activity. VVB has assessed the TEM Simi Oa contract.pdf /39/ which establishes this relationship between the two entities. The contact details and role of this entity have been specified in the PD. An end user agreement/54/ will be signed by each participating

household confirming the ownership rights of the carbon credits generated from this project belong solely to the project proponent. Earthood has checked and assessed the blueprint of the End User Agreement/54/ which will be provided to the end-users and found it appropriate.

2.2.5 Project Type

The project is a non-AFOLU, group project. The project is dealing with the energy efficiency sector primarily. The validation team has assessed the project and confirms that it contributes towards 8 SDG goals which are 1, 3, 4, 5, 7, 8, 13 and 15.

SD Vista Sectoral Scopes	U.N Sustainable Development Goals
1. Agriculture, forestry and other land use	Life on Land 15.2.1
2. Climate Change Adaptation	Climate Action 13.0
3. Education	Quality education Goal 4.1 Goal 4.2
4. Energy	Affordable and clean energy Goal 7
7. Health	Good health and well-being Goal 3
10. Livelihoods	No poverty Goal 1 Decent work and economic growth Goal 8 8.5.1
14. Women's empowerment	Gender Equality 5.4.1 Goal 5

2.2.6 Project Location

The location of this project activity is the geographical boundary of Papua New Guinea (PNG), the same has been verified through the google earth and is found to be appropriate. Also, the household chosen for pilot study/30/ is found to be within the geographical boundary of PNG. Location of New PIs will be recorded in the database and checked by the VVB during verification.

The four provinces involved as the first project instances of the group activity in Papua New Guinea.

S.No.	Location	Latitude	Longitude
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1.	Southern Highlands Province	6.4179° S	143.5636° E
2.	Eastern Highlands Province	6.5862° S	145.6690° E
3.	Western Highlands Province	5.8310° S	144.1720° E
4.	Jiwaka Province	5.8564° S	144.5218° E

This was found to be acceptable by the VVB.

2.2.7 Baseline Scenario

In the absence of project activity, the baseline scenario is that the target population continues to consume non-renewable biomass fuel to meet similar thermal energy needs as provided by project ICS cookstoves. The target population, prior to project activity, also exhibited multidimensional poverty, with indoor air pollution arising from traditional cookstoves also making the female population particularly susceptible to respiratory ailments. This has been validated from results. In addition to this the time spent on gathering fuel for basic thermal energy needs daily, also depletes children and women from pursuing other useful activities. This leads to low education among the population.

The same has been assessed from the baseline study conducted by a third-party consultant; report of baseline survey has been shared by the PP. The study conducted has been further supported by published literature like ACIAR, 2013, IRENA report of 2013 and Page et al. (2016)/44, 53, 48/, which has been checked and further validated by interviewing the end users during the remote audit /7/11/.

Social Capital

PP has demonstrated baseline scenario in terms of the social conditions of the host country, it has been checked through the UNICEF data/57/ that 40% of population in PNG is living below the extreme poverty line and 41% of children live in poverty. It has been confirmed that 80% of the population in PNG doesn't have basic services like electrification, vaccinations, or basic improved cookstoves, according to UN Habitat data/58/. PP has conducted the baseline study/38/ with 220 households, that demonstrates that 98% of the households uses firewood and 2% of the households uses electricity or gas as a main source of energy, which has been further supported by published literature like ACIAR, 2013/44/, IRENA report of 2013/53/. As per WHO/59/, air quality in PNG is moderately unsafe, where indoor air pollution and smoke generated from burning of solid fuels is linked to illness and death.

Economic capital

PP has demonstrated the economic conditions of the host country, through the world bank data on employment. It has been confirmed that PNG accounts for 44.2% in vulnerable employment and Informal sector (which have the lower income than the formal jobs) is popular among the youths. Other studies like Hanson, et al., 2001/42/ says 40% of the population living near the poverty line with average income as K439/ Month as per Bonilla, et al., 2020/60/.

Natural capital

PP has demonstrated baseline scenario in terms of natural conditions of the host country. PNG has high biodiversity with a total area of 462,840 km², with forest covering about 65% of the land. It has been confirmed through the WWF data/61/, that from 2002 to 2023 PNG has lost about 943,00 ha of the humid primary forest, which makes a total of 51% forest lost for the same period.

The data and sources used by the PP, clearly depicts the social, economic and natural capital of the host country, and is found to be authentic and reliable. The procedure for identifying the baseline

scenario has been correctly followed and the identified scenario reasonably represents what would have occurred in the absence of the project. VVB confirms that the assumption and data can be deemed reasonable.

2.2.8 Causal Chain(s)

The PP has presented the causal chain in the PD for the activities performed under the program, identifying effect, outcome and impacts, on the plants and people and their prosperity. The project includes nine impacts as identified by the PP, which have been categorized in the Positive impacts (8 positive) and negative impacts (1 Negative). The Validation team assessed the causal chain mentioned in section of 2.1.9 of SD VISTA PDD through remote audit interview and supportive documents submitted by PP.

Following documents are assessed by the validation team to validate the causal chain:

SD VISTA PD/05/

VCS Final Validation Report /22/

Project database/30

Baseline assessment report/16/

Ex-ante ER sheet/06/

Manufacturer's Specifications /51/

Training material and training attendance records/32, 35/

End-user Carbon Waiver agreement. /54/

Anti Bribery and Corruption Policy /10/

Employee Handbook /26/

Modern slavery Policy of PP/14/

Grievance Redressal Mechanism/31/

Peer Reviewed Scientific Papers/48, 49, 50/

Pilot Study conducted by TEM (report prepared by Simi Oa/36/

VVB has reviewed all the causal chain presented in appendix A of the SDVista PD. PP has appropriately and comprehensively described the causal chain in the PD, correctly tracing the positive and negative, intended and unintended impacts on the planet and on the people and prosperity, as required by para 2.1.5 of the SDVista Standard/1/. The validation team confirms that the PP has included all reasonable effects of the project activities in the causal chains. No SD VISTA assets have been generated by the project.

2.2.9 Threats to the Project

The steps taken to assess the likely natural and human-induced threats identified by the project to the expected sustainable development benefits during the project lifetime are discussed below:

Nature-induced Threats

Threats	Solution	Conclusion
Continued deforestation and degradation make firewood an untenable source of biomass.	As much as the efficiency of the cookstoves coupled with the imminent scarcity of the wood fuel would support the use of G3 Burn rocket stove (also known as Kuniokoa Cookstoves) as proposed by Tasman Environmental Markets Asia Pacific for this project, there is the potential that as the population continues to expand and there may be a point when firewood becomes an untenable source. In response to this, Tasman Environmental Markets in country team will train end users to maximise the use of small twigs and branches, crop residues near the local communities they live.	As confirmed by the assessment team during the remote audit interviews with end users /11/ and the training manual /35/, the end-users will be trained to make the best use of the resources available in the locality and hence, the threat is negligible.
Human-induced threats		
Stove users' lack of upkeep of stove resulting in either malfunctioning of the stove as designed or lack of stove use.	Tasman Environmental Markets Asia Pacific will conduct routine visits to end-user households at least twice a year in each main geographic areas that currently use the new G3 Burn rocket stove (also known as Kuniokoa), to identify potential defects in the stove's construction, maintenance, and sub-optimal use. To reduce this proposed treat, Tasman Environmental Markets Asia Pacific's in-country team will hire, train, and supervise users that will involve visiting the end-user's houses twice per year to monitor the use of the new ICS.	The assessment team during the interviews/11/ with the end-users received positive feedback regards with TEM's grievance redressal mechanism. End Users responded positively when asked about the after-sale care provided by TEM. It is also concluded from interview with beneficiary that TEM representatives have carried out awareness exercises to inform HOUSEHOLD of working and maintenance of the stove. Interviews with the on-ground team also confirmed their robust assessment of stoves conducted regularly. Assessment of Training Records and Training material/32, 35/ by VVB confirms the skill and knowledge of the on-ground team regarding the maintenance and functioning of the ICS. It is concluded from remote audit interviews that PP conducted routine spot audits to detect defects and issues in the stove maintenance and functioning.

Failure of Behaviour Change resulting in high levels of non-adoption	Tasman Environmental Markets Asia Pacific will continuously research the impacts of behaviour change for successful adoption. Study outcomes, whether formal or informal, will be included in future training and community engagements to continue shaping knowledge, and enhancing positive peer-to-peer influence. Additionally, non-adopters will also be engaged to understand the potential barriers preventing them to adopt the use of the ICS. These outcomes and conclusions will be useful to improve future training events with the goal of to avoid miss-information. To add to the above, Tasman Environmental Markets Asia Pacific will utilize Equation 6 of the Methodology, under which the μ_y parameter for the Bold,adjusted parameter is listed as 0.1 or 10%. Signifying the fact that 10% of end-users may potentially use both ICS as well as open fire cooking practices.	It is confirmed during remote audit interviews and desk review that the PP the awareness measures as proposed by the PP are robust, reasonable and implementable. The interviewees were positive in their approach towards the adoption of ICS as it is more efficient and less polluting /11/ Records from LSC/25/ confirm that information is mediated to beneficiaries in their local language. The PP has also taken into account the fact that 10% of end-users may potentially use both ICS as well as open fire cooking practices and imbibed the same in their ER calculations.
ICS Projects and distributions are not shown to have an increased impact on household uptake.	Co-benefits like time savings and increased health benefits aligned with the adoption of ICS work as prominent incentives for increased uptake of these stoves. In response to this, Tasman Environmental Markets Asia Pacific will conduct trainings highlighting how time saving, health benefits, previous users' positive experiences are important factors for large scale adoption	The assessment team concluded from a interviews with pilot beneficiaries that TEM representatives have carried out awareness exercises to inform HOUSEHOLD about the multiple benefits that the stove brings. The interviewees were positive in their approach towards the adoption go ICS as it is more efficient and less polluting /11/

2.2.10 Benefit Permanence

PP has planned several interventions throughout the crediting period of the project activity to bring out the behavioural change in the intended community of the host country to shift to the use of the cleaner cooking practices.

The following documents are assessed by the validation team to validate the project's benefit after project activities cease.

- Project database/30/
- Training material and records/32, 35/
- End User Agreement and Proof of start date /54/
- Stakeholder Consultation Meetings/15/25/
- Interviews with stakeholders and end users during remote audit/11/
- Literature Review /44, 48, 49, 50, 53/
- TEM's MoU with Western Pacific University (WPU) /55/

In the project description /05/, it is stated that to ensure the long-term sustainability of the ICS project and the lasting advantages of ICS and other program benefits, PP has committed to the following long-term objectives:

1. Regular and periodic monitoring will be conducted by spot-audit teams of the PP to ensure the use of ICS by the end-users. The teams will be provided with transportation (bicycles, motorbikes, or motor vehicles) and communication (smart phones) facilities. This information was confirmed by the assessment team during their remote audit /7/
2. Regular trainings regarding the long-term environmental, health, nutritive, and resiliency benefits that come with the adoption and usage of ICS will be conducted for locals. This information was confirmed by the assessment team during their remote audit /7/ and through additional sources such as training session records, presentations, and LSC records /25/.
3. TEM has forged a partnership /55/ with Western Pacific University (WPU) in Papua New Guinea to develop sustainability related courses for their students. The PP has also specified that it envisages to continue working with rural communities through periodic monitoring, training sessions and partnering with potential beneficiaries such as WPU to further secure the permanence of the behavioural changes achieved through the project activity.

Apart from this PP has also developed a continuous grievance redressal mechanism which continuously addresses the challenges associated with distribution and use of ICS or any other concern associated with the project activity to foster trust and sense of reliability amongst beneficiaries. TEM also aims to provide training/32/ to locally recruited staff which ensures development of employable skills, thus ensuring their ability to secure livelihood even when the project ceases. The stated intended outcome of the benefit permanence has been reviewed and found to be appropriately stated for all the activities planned. The indicated permanence strategies are found to be appropriate by the VVB to increase the awareness of the climate change and change the behaviour of the communities over the project lifetime.

The assessment team confirmed the benefit-permanence activities through a desk review and site visit interviews with project proponent, and they believe that the measures will likely help the project achieve its long-term sustainability goals.

2.3 Stakeholder Engagement

2.3.1 Stakeholder Identification

To assess the process of stakeholder's identification, following steps are taken by the assessment team:

1. Review of SD VISta PD/05/.
2. Section 2.2 – Sustainable Development Verified Impact Standard/01/
3. Local Stakeholder consultation Report/15/.
4. End user agreements/54/.
5. Invitation Emails, Newspaper Advertisement, and List of attendees/24, 25/.
6. Remote audit by the validation team and interviews with PP and the local stakeholders/7/11/.

In the first stage, PP has identified the benefits and co-benefits from the project activities. This stage is found to be appropriate since this helps in identifying the affecting stakeholders based on the impact and outcome of the project activity. In the second stage, PP has engagements with the project implementation partners (Simi Oa Consultancy Services) to better understand the communities of the host country. PP has submitted an agreement with the Simi Oa Consultancy Services/39/ as evidence. Alongside, PP has taken approvals from the local authorities like Provincial Government of the Southern Highlands Province to share the information about the project and have their feedback and support for the implementation of the project, which has been checked and is found to be appropriate. PP has also informed local NGO 'Mt Giluwe Conservation Foundation' about the project activity, which has been reviewed through the invitation letter sent to the NGO/62/. The stakeholders have been identified based on the directly and indirectly affected groups. In the third stage, PP conducted the local stakeholder consultation (details of which has been provided in section 2.3.3) and baseline study in the host country. The PP has shared the evidence for contacting the Provincial Governments and receiving the approval letters for pilot study from those governments/64/.

The process of stakeholder identification has been reviewed and found to be sufficient to identify the direct or indirectly affected group.

Finally, three categories of stakeholders have been identified, which includes

1. Local people, communities and or representatives who are directly affected (Directly affected Group),
2. National governmental officials or National Focal bodies (Indirectly affected Group), and
3. Local NGOs or Women Group (Indirectly affected Group)

The Validation team via desk review and interview with stakeholders (including end users, community leaders, local stakeholder participants) confirms that the direct and indirect stakeholders identified are appropriate and comprehensive, wherein no potential stakeholder is left out of the purview.

2.3.2 Stakeholder Description

As assessed in the previous section, the stakeholders identified in above section 2.3.1 and the means of validating them are described below:

Current Beneficiaries: Any family/individual/ woman who is involved in cooking using traditional cookstoves and shows interest in adopting ICS is referred as current beneficiary. Family members/ people residing in the same household compound are referred to as beneficiary families. VVB assessment team during remote audit/7/11/ interviewed such families and found that they had been given information about the benefits of ICS and were willing to adopt it.

Potential Beneficiaries: While Current and Potential beneficiaries have similar characteristics, the primary difference arises from lack of awareness of lack or financial capability to afford ICS. Tasman Environmental Market Asia Pacific conducts awareness programs to engage more people in Project Activity. The same has been assessed during the remote audit/7/11/ and by stakeholder consultation evidence provided by PP./30/15/

Local Vendors: This refers to those who are involved in supplying fuelwood to the project area. The PP identifies probable negative impact and has chalked out ways to mitigate the same. The same has been elaborated in Section 3.2.

Local NGO and Business and Government Authorities: Tasman Environmental Market Asia Pacific works with local country partner for implementation and management of PA, while also ensuring that the local government is taken into consideration throughout all phases of the Project Activity. To assess the same, assessment team checked all the authorisations and NOLs received from different provincial government officials and departments (Western Highlands Province, Southern Highlands Province, Eastern Highlands Province, Jiwaka Province, PNG-Forest Authority). PP has also shared the invitation letter/62/ and letter of support/63/ received from the Local NGO 'Mt Giluwe Conservation Foundation' for the LSC meeting.

Media and Academic Institutions: Tasman Environmental Market Asia Pacific involves with media to spread awareness, and is also open to collaborate with institutions to academically study and understand the PA. The Baseline assessment report has been conducted by Tasman Environmental Market Asia Pacific in collaboration with Simi Oa Consultancy Services /16/, hence engaging academia involved in policy research.

The validation team verified the information from the end user agreements for the pilot project/54/ signed by all the end users shared by PP and the newspaper advertisements, stakeholder meeting recordings and the list of attendees/24, 25/.

2.3.3 Stakeholder Consultation

The local stakeholder consultation process has been described under Section 2.2.3 of the SD VISTA PD/05/. The Project Participant correctly identified the relevant stakeholders such as local NGOs, representatives of local authorities, target project ICS users. The local stakeholder meeting was conducted from 13/03/2023 to 20/03/2023.

S/N	Province	Date
1	Southern Highlands Province	13/03/2023
2	Western Highlands Province	15/03/2023
3	Jiwaka Province	17/03/2023

4	Eastern Highlands Province	20/03/2023
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The participants across the host country (Papua New Guinea) were invited and informed about the meeting through email invite, newspaper advertisement, radio broadcasting, invitation to churches and Facebook/social media announcement notifying the date, time, and location of the event. The invitation evidence has been provided by the PP and is found to be appropriate. Various governmental officials and Local NGO 'Mt Giluwe Conservation Foundation' has been invited to the local stakeholder consultation, which has been verified through the invitation letter/62/. The attendance sheet, LSC Report, photos and videos of the local stakeholder meetings were submitted as evidence by the PP/25/. As evident from these documents and the remote audit interviews /7, 11/, the consultation was carried out with due consideration and respect for local customs, values, and institutions. People of various economic groups and genders following the local culture were observed participating in the meeting. The participation was checked through the photographs of the LSC/25/ and attendance sheets/25/. Women participated in large numbers in all the LSCs that took place for this project activity.

All questions raised by stakeholders were answered and documented by the PP and their representatives. Stakeholders had no major comments/complaints/grievances by the end of the meeting which could have any significant modification in the project description or its design. Therefore, the project design and implementation has not been revised with this regard. The date and mode of invitation were confirmed from remote interviews of the LSC participants conducted by validation team. VVB validation team was also able to confirm through the interviewed stakeholders that they were part of the LSC meetings conducted by the PP and were found to be aware about the proceedings of the meeting including benefits of the improved cookstove, including lesser smoke, lesser time in wood collection, lesser wood consumption, etc. This was also confirmed through the feedback forms/23/ by the LSC participants. During the remote audit conducted by the assessment team, the VVB confirmed that there was no language barrier for the local stakeholders as the project manager communicated in tok pidgin, the local language used in PNG. The same has been verified from the LSC report /25/ shared by the PP. Locals were invited through newspaper and radio advertisements /24/, to compensate for the poor internet connectivity in a few regions. As per the VVB's assessment, PP has addressed all questions and responses of all the stakeholders with satisfaction and for future grievances PP has provided the communication details that can be used for placing the complaints after the implementation of the project.

The PP has a continuous grievance mechanism in place for the local stakeholders. The beneficiaries of the project ICS have been provided with the following means to contact the PP:

- Directly registering complaints with the in-country team of the PP. Emails IDs and phone number of the in-country Project Manager: Michael Kisombo, have also been provided
- A toll-free helpline on which the end users can seek assistance.
- E-mail at pngcookstoves@tem.com.au
- Reach out to TEM via the project's website, <https://www.tasmanenvironmental.com.au/pngcookstoves/>

Evidence on the process of collection of grievance feedback from the end-users is provided by PP, thus the requirement is identified as being fulfilled.

2.3.4 Continued Consultation and Adaptive Management

The following steps have been taken to assess the project's plan for continuing consultation and adaptive management with the stakeholders.

Review of SD VISTA PD/05/.

Local Stakeholder consultation Report/15/.

Invitation Emails, Newspaper Advertisement, and Invitation List/24/.

Tasman Environmental Market Asia Pacific Website announcement /27/

Remote audit by validation team and Interviews with PP and stakeholders/7/11/

List of Attendees and feedback providers/25/

Supplementary photos of LSC Meeting/25/

Filled Stakeholder Evaluation Forms/23/.

Grievance Redressal Mechanism/31/

The assessment team validated that the project plan to engage in continued stakeholder consultation and adaptive management of the project, as described in Section 2.2.4 of the SD VISTA PD. The information given in SD VISTA PD is found correct and appropriate with the information given in the supportive evidence. The PP mentions that quarterly feedback is taken from stakeholders by the sales visit. The same has been validated through remote audit.

2.3.5 Anti-Discrimination

The validation team checked and confirmed that Tasman Environmental Market Asia Pacific does not discriminate based on race, age, colour, sex, national origin, physical or mental disability, or religion. PP have TEM's Employee Handbook/26/ which is validated during validation process and during remote audit/7/ interviews with the workers it is confirmed that no discrimination based on race, age, colour, sex, national origin, physical or mental disability, or religion by the PP.

2.3.6 Worker Training

The PP plans to conduct yearly learning and training sessions for employees who are recruited from the local community. Interviews during the remote audit confirmed employees were trained and well-versed in the skills needed to carry out their jobs.

The assessment team validate the information from several sources, which are:

1. Training Records/pictures/32/
2. Training coursework- PPTs/35/

The audit team deems that the project has properly identified the training needs and delivered adequate capacitation to project's workers for them to perform their activities in a safe and effective manner.

2.3.7 Equal Work Opportunities

The validation team assessed and confirmed that Tasman Environmental Market Asia Pacific provides equal employment and advancement opportunity to all regardless of age, gender identity, sex or sexual orientation, race, religion or belief, marriage and civil partnership, pregnancy or maternity, disability, or any other legally protected category which is mentioned in the statement of policy for Equal opportunities and anti-discrimination in Section 14 of the TEM 2023 Employee Handbook /26/ which was validated during validation process and during remote audit/7/ interviews with the workers it is confirmed that equal work opportunities are provided to all the workers. Further the Employee

Handbook /26/ was thoroughly assessed, wherein strict repercussions of any instance of discrimination amongst the workforce is noted, to further validate the above.

2.3.8 Workers' Rights

The project activity is in compliance with laws including national, sub-national, state, and local, as well as policies, laws, rules, and regulations.

Since PNG is a part of ILO convention, the project activity is also in accordance with the terms and conditions as per Core Labor Conventions of the International Labor Organization (ILO) which has been validated during desk review.

The assessment team also assessed TEM's Staff Handbook/26/ and found it to be in accordance with the PNG labour legislation /45/. As per the PD, the PP plans to have a labour contract with each in-country personnel following PNG labour legislation. It will be explained explicitly before the workers sign it.

The assessment team also carried out literature review to assess the same. The team referred to scientific writings/48, 49, 50/ and came to the same conclusion as established above.

2.3.9 Occupational Safety Assessment

PP has provided potential occupational safety hazards, and mitigation measures. During the remote site visit, PP representatives and surveyors were interviewed, and have been informed of risks and instructed how to minimize them. It should be noticed that use of stove does not increase the safety risks that are present in the landscape of the project. PP will be minimizing the occupational risks through community familiarity, language fluency, and being native to the culture. Hence, in assessment teams' opinion occupational risks are comprehensive, and measures has been put in place to minimize risks to workers.

2.3.10 Feedback and Grievance Redress Procedure

PP has designed a feedback and grievance redress process, where the stakeholders and end-users will have the mail and contact information and understanding that they should contact the organization for any problem, or grievance. PP has discussed the same during the local stakeholder consultation as well. Various channels for the grievance reporting are Physical walk-in, Social media and online, phone call and verbal through community representative.

The grievance process described has been reviewed through TEM Grievance Redress mechanism/31/, dated July 2023, and found to be in line with the document. The grievance mechanism has integrated traditional methods for the conflict resolution. PP has different channels to report grievances which includes physical walk in, social media and online, phone call, verbal communication through community representative. The procedure acknowledges and prioritizes traditional and customary conflict resolution methods in Melanesian society by promoting dialogue facilitated by clan leaders and community-level staff. It also employs Field Assistants from local communities to serve as first responders in addressing grievances, which is found to be appropriate. Detailed description on the reporting channels and process along with the stipulated timelines has been provided in SDVista PD/05/. Resolution may be complex and time and resource intensive, wherein it is referred field coordinator to Implementation Partner to operation manager to company lead to head office, depending on complexity. During the remote audit, end-users and stakeholders interviewed, know the process and the contact information of the team, where they can contact for any grievance. The above two processes are validated via interviews conducted with end users and the coordinators during remote audit. /7/. VVB has assessed the above by checking functionality of the provided contact numbers and checking presence of grievance records at site office during RSV.

2.3.11 Feedback and Grievance Redress Procedure Accessibility

PP has a continuous grievance mechanism in place for the local stakeholders. The beneficiaries of the project ICS have been provided with the following means to contact the PP:

- Contacting in-country Team during the twice a year visits to maintain the ICSs. The contact details of Tasman Environmental Markets Asia Pacific in-country Project Manager: Michael Kisombo, have also been provided. (+675 7262 3864, mkisombo@hotmail.com and currently michael.kisombo@tem.com.au) The in-country Project Manager holds the Grievance register for the project activity.
- A dedicated email id (pngcookstoves@tem.com.au) on which the end users can seek assistance.
- Reach out to TEM via the project's website, <https://www.tasmanenvironmental.com.au/pngcookstoves/>

The details were confirmed during the remote audit marked as visit/7/. The PP has provided accessible channels for registering complaints, including a telephone number and an email address. Both these communication methods have been verified and found to be operational.

2.3.12 Stakeholder Access to Project Documentation

Section 2.2.12 of SD VISta PD mentions that upon the household's receipt of the ICS, the PP will inform the beneficiaries and their respective families about their participation in a project that distributes the G3 Burn rocket stove. The Project Design Document is also uploaded on the VERRA website. The presence of the same was confirmed by VVB. The PP plans to publish it on its website (<https://www.tasmanenvironmental.com.au/pngcookstoves/>) during the crediting period.

2.3.13 Information to Stakeholders on Validation and Verification Process

The PP aims to notify the end users and their families for the assessor's visit in a timely manner before the site visit occurs (at least 7 days through the website linkage, phone, email or in person), They will also be made aware of ICS generated carbon credits which in turn are used to cover the cost of ICS production and distribution through End User Warranty Card and agreements/54/.

The individuals throughout PNG who took part in the event were notified about the meeting via email invitations, newspaper ads, and announcements on Facebook and social media, word of mouth. These notifications included details about the event's date, time, and venue. The PP submitted the attendance records /25/, along with photos and videos from the local stakeholder meetings, as supporting evidence. /25/

Further it was confirmed that the PA was in compliance with paragraph 3.5.5 of the SD VISta program guide/2.1/ which states that, "The project proponent shall also provide notification of any upcoming assessor site visits to potentially affected stakeholders, preferably with 30 days' notice. Such notification shall indicate the assessor's name, assessment team leader, dates, and locations of the audit, contact details, and means of communicating with the assessment team." The end user had been intimated of all the above pointers by local, on ground staff, 30 days in advance to the assessment team's visit in May 2023. This was verified via interviews with various stakeholders/11/.

2.4 Project Management

2.4.1 Avoidance of Corruption

As per the SD VISta PD /05/, Tasman Environmental Markets Asia Pacific has provided details of the anti-corruption regarding the project and in general. Tasman Environmental Markets Asia Pacific's policy /10/ in this regard was assessed and confirmed further during the interviews with the staff of Tasman Environmental Markets Asia Pacific, its associates and end users. No evidence of any form of corruption or illegality was found during the same.

The PP has specific Employee rules which insist a strong punishment up to criminal prosecution / termination of job for the bribery / forgery related offences. This has been verified from TEM's Employee Handbook. /26/

The audit team believes that the project management has created and established a solid and thorough framework to prevent the commission of any form of illicit acts by project proponent after evaluating all the aforementioned policies. During desk review and remote audit, there was no indication of any kind of corruption or unlawful activity conducted in the project activity.

2.4.2 Statutory and Customary Rights

In SD VISta PD, section 2.3.2 Statutory and Customary Rights mentioned that the project involves distribution of ICS to individual households only and it will not involve any land use or acquisition. It is validated by Earthood assessment team, and the agreement of pilot end user /54/ mentioned clearly about the rights of the PP in the project and the same is verified during the remote audit/7/.

2.4.3 Recognition of Property Rights

SD VISta PD, in section 2.3.3, mentions that ICS were distributed free of cost to end users and all property rights of ICS belong to end users, while carbon credit generated by the project belongs to Tasman Environmental Markets Asia Pacific which was assessed via Warranty card and end user agreement/54/. The same has also been confirmed during the remote audit/7/ that no property rights were transferred by the end users to the PP.

2.4.4 Free, Prior and Informed Consent

In SD VISta PD section 2.3.4 'Free, Prior and Informed Consent' mentions that it is a voluntary project. End users are free to choose whether they want ICS or not. Agreement with End User/54/ are signed between the end user and PP if he wants to take part in the project. The validation team confirms the information during desk review and remote audit interview with beneficiaries who confirmed that they were participating in the project with free will, without any outside coercion.

2.4.5 Restitution and/or Compensation for Affected Resources

The validation team confirmed during the remote audit that the project is solely involved in distribution of ICSs to individual households only and it will not affect any resources, natural or otherwise. Hence there arises no necessity for restitution or compensation of any kind.

2.4.6 Property Rights Removal/Relocation of Property Rights Holders

The validation team confirmed that the project is not involved in any removal of property rights or relocation of property rights holders. The project involves distribution of ICSs to individual households only and it will not involve any land use or acquisition. It is validated by Earthood assessment team, and the agreement of End User/54/ mentioned clearly about the rights of the PP and End Users in the project and the same is confirmed during the remote audit/7/11/.

2.4.7 Identification of Illegal Activities

In SD VISta PD section 2.3.7 Identification of Illegal Activities mentioned that there are no illegal activities found which could affect the product activity. The validation team confirmed this during remote audit interviews. Avoidance of corruption is already discussed in section 2.4.1 of this report. Tasman Environmental Markets Asia Pacific has a stringent group policy for preventing the discrimination in any form and does not tolerate any form of unlawful discrimination. This has been verified from Section 5.4 of TEM's Employee Handbook. /26/. VVB assessment team did not find any instance of unlawful activities including corruption and discrimination during the course of the validation process.

2.4.8 Ongoing Conflicts or Disputes

In SD VISta PD section 2.3.8 Ongoing Conflicts or Disputes, mentions the inaptness of the issue, given the installation of ICS does not affect property right. It is to be noted that there is no ongoing or unresolved conflicts or disputes over rights to lands, territories and resources and any disputes that were resolved during the last twenty years and validation team has not found any evidence of ongoing conflicts or disputes.

2.4.9 National and Local Laws and Regulations

The Earthhood validation team did not detect any incompliances related to laws, statutes, and regulatory frameworks about the application of improved cookstoves in PNG. It validated during desk review, remote audit interview/7/ and through web search that Tasman Environmental Markets Asia Pacific and its local partners obey all relevant local laws and regulations of PNG. Additionally, PP has provided the letter of confirmation of PNG Forest Authority/47/ and letter of confirmation from the local authorities/64/. Compliance with the following laws, rules and frameworks have been demonstrated in the PD. The VVB has reviewed the same and found the compliance in place:

International Laws, Regulations and Conventions

- United Nations Framework Convention on Climate Change (UNFCCC,1992)
- United Nations Declaration on the Rights of Indigenous Peoples Rights (2007)
- Stockholm Declaration of the United Nations Conference in Human Environment (1972)
- Rio Declaration on Environment and Development (1992)
- Paris Agreement (2015)

Papua New Guinea's Laws, Regulations and Conventions

- Papua New Guinea's Constitution (1975)
- Climate Change Management Act 2015, Amended (2022), and proposed Carbon Market draft regulation (2023)
- Papua New Guinea Vision (2050)
- Workers Compensation Act (1978)
- Discriminatory Practices Act (1963)
- Child Welfare Act (1961)
- Employment Act No. 54 of 1978 (Consolidated up to 31 March 2001)
- Industrial Health Safety and Welfare Act (1962)

- Industrial Relations Act (1962)
- Investment Promotion Act (1992)
- Environmental Act (2000)
- Forestry Act (1991)
- Climate Change Management Act 2015, Amended (2022), and proposed Carbon Market draft regulation (2023)

The initiative is voluntary, as PNG lacks a mandate for ICS distribution. Additionally, the project falls outside the scope of the Environmental Impact Assessment (EIA) under PNG laws and regulations.

An assessment of Tasman Environmental Markets Asia Pacific's Policies validates its conformance to all local laws and regulations.

2.4.10 Project Ownership

Tasman Environmental Market Asia Pacific is the project proponent. The project device, i.e., G3 Burn rocket stove, is distributed free of cost to end users and is the property of the beneficiary. This has been confirmed via assessment of end-user agreement/54/ and interview with PP representatives and end-users/11/ during remote audit. An end user agreement/54/ has been signed by each participating household confirming the ownership rights of the carbon credits generated from this project belong solely to the project proponent. Earthood has checked the End User Agreement/54/ and assessed the same.

2.4.11 Grouped Projects

The project is a grouped project activity. It is about distribution of energy efficient stoves for household cooking purpose in Papua New Guinea. Eligibility criteria for all new Group project activity instances to be added during the crediting period are specified in Section G1.1 of the Sustainable Development Verified Impact Standard/01/.

Section 2.3.11 of the SD VISta PD /05/, establishes the necessary eligibility criteria to ensure that inclusion of all new project activity instances is in accordance with the requirements detailed in the Sustainable Development Verified Impact Standard /01/.

3 BENEFITS FOR PEOPLE AND PROSPERITY

3.1.1 Condition of Stakeholders at Project Start

SD VISta PD section 3.1 describes the Condition of Stakeholders at Project Start. PP has adequately explained the circumstances of the stakeholders in PD, who are described as Beneficiaries (end users), Local Vendors, Local NGOs and Business, Government Authority, Media, and Academic institutions in the host country; all the stakeholders are described in detail. The validation team has checked various literature including World bank data to access the economic conditions in the baseline, UNICEF data/57/, UN Habitat data/58/, ACIAR, 2013/44/, IRENA report of 2013/53/, WHO data/59/, Hanson, et al., 2001/42/, Bonilla, et al., 2020/60/, and WWF data/61/, the information provided is found to through authentic source and hence, acceptable.

The diversity aspects of stakeholders and the changes in the same have been assessed. No significant changes in pattern have been observed in the baseline social (geographical locations, rural population, opportunities), Economic (Unemployment rates, income levels) and cultural (religion, language, cooking patterns) conditions of the end users in the recent past prior to PA implementation. Similarly, no significant patterns of change have been observed in the conditions of the implementation partners and government representatives in the years preceding PA.

Validation team has confirmed the information during the remote audit interview with different stakeholders, including representatives of PP, beneficiaries, leaders of local community, , and found all the information in PD to be accurate. VVB has further corroborated remote-site observations with results of studies conducted during the Baseline Survey /16/ conducted by PP in October-November 2022 and other peer reviewed papers on the subject. /48, 49, 50/

3.1.2 Expected Stakeholder Impacts

Section 3.2 of the SD VISta identifies the expected stakeholder impacts of project activity. The following table summarizes the means used to assess the expected impacts on each stakeholder group resulting from project activities:

Impact #1	Access to Improved Cooking Technology
Type of Impact	Positive, actual, direct
Affected Stakeholder Group(s)	Beneficiaries & beneficiaries' families
Resulting Change in Well-being	The project activity aims to replace and estimate 100,000 traditional cookstoves with fuel efficient improved cookstoves. Upon gaining access to the ICS, inadequate cooking will be gradually reduced. As will the burden of firewood collection, the time to prepare food and the cost to buy firewood, hence improving health and well-being.

VVB Assessment	The validation team confirmed the information from the document review, sectoral expertise and remote audit interviews with various stakeholders including PP, end users, community leaders , confirms the expected impact of this project on the Affected Stakeholder Group(s). The description in section 3.2 of the SD Vista PD /05/ is deemed appropriate, correct, and feasible.
Impact #2	Improved Health Status
Type of Impact	Positive, predicted, direct
Affected Stakeholder Group(s)	ICS end-users
Resulting Change in Well-being	Prior to the project activity, high levels of smoke in the cooking place were observed during the pilot study. Through the use of the ICS, gradually reduce the local people's exposure to high PM2.5 and high CO due to higher efficiency of combustion within their household over the lifetime of the project activity. The monitored baseline emission level is 943.3 mg/MJ, and the project aims to monitor a PM2.5 value lower than the baseline during the project activity through a sampling process during the monitoring period of the project.
VVB Assessment	The validation team confirmed the information from the document review, sectoral expertise and remote audit interviews with various stakeholders including PP, end users, community leaders, confirms the expected impact of this project on the Affected Stakeholder Group(s). It was validated from remote audit interviews /11/ that smoke has reduced indoors with the usage of ICS. The description in section 3.2 of the SD Vista PD /05/ is deemed appropriate, correct, and feasible.
Impact #3	Reduction of time spent on unpaid domestic work
Type of Impact	Positive, Predicted, Direct
Affected Stakeholder Group(s)	Beneficiaries (most notably, female, elderly, and children, primarily girls)

Resulting Change in Well-being	Due to higher thermal efficiency, less firewood is required during the cooking process, and time spent cooking is significantly reduced. This results in a direct reduction in time spent on firewood collection and cooking for people. In the baseline, it was found that end-users would collect firewood 3-5 times/week. Upon sampling during the monitoring period, this frequency of collection should reduce as less firewood is required for cooking and heating throughout the week.
VVB Assessment	The validation team confirmed the information from the document review, sectoral expertise, and remote audit interviews. Conversations with female end users about the perceived impact of ICS on women stated lack of free time as a factor hampering entrepreneurial ambitions in women and education in girls. The description in section 3.2 of the SD Vista PD /05/ is deemed appropriate and correct

Impact #4	Increased Income
Type of Impact	Positive impact and increase employment opportunity are predicted with the project activity
Affected Stakeholder Group(s)	Local Citizens, Employees hired for Project Activity
Resulting Change in Well-being	During the project crediting period, Tasman Environmental Markets Asia Pacific will oversee maintenance of cookstoves and the monitoring plan, which will require the need to hire local people. Local people will be hired through a fair and unbiased process, during which we estimate that 130 jobs (10 full-time and 120 part-time) will be created over the project lifespan. Employees will be compensated above the average income of K439/month.

VVB Assessment	The validation team confirmed the information from the document review, sectoral expertise, and remote audit interviews with stakeholders. Assessment of employee handbook, and TEM's Policy and interview with current employees confirms the current and further future employment creation. The increased income will be checked at the time of verification through the employment records and employee payslips. This confirms the expected impact of this project on the Affected Stakeholder Group(s). The description in section 3.2 of the SD Vista PD /05/ is thus deemed appropriate and correct.
Impact #5	Trainings imparted on climate change, project implementation and monitoring procedures
Type of Impact	Positive, Predicted, Indirect
Affected Stakeholder Group(s)	Local community, Beneficiaries, TEM-PNG in-country team and Staff
Resulting Change in Well-being	Prior to the project activity, local residents and end-users had little awareness of ICSs and climate change. With the inception of the project activity, training and skill development related to community engagement, survey conduction, technical trainings like conducting Water Boiling Tests (WBT) will be provided to many stakeholder groups which is envisaged to empower their lives by not only providing long-term employability but also through increased awareness levels regarding issues related to climate change, social equity. This will be done during the onboarding of new staff throughout the lifespan of the project activity.
VVB Assessment	Training and skill development related to community engagement, survey conduction, technical trainings including those for conducting Water Boiling Tests (WBT) will be provided to many stakeholder groups which is envisaged to empower their lives by not only providing long-term employability but also through increased awareness levels regarding issues related to climate change, social equity. Interviews with current employees confirm that they have received adequate training. The description in section 3.2 of the SD Vista PD /05/ is deemed appropriate, correct, and feasible.
Impact #6	Improved Economic Outcomes
Type of Impact	Positive, actual, direct

Affected Stakeholder Group(s)	ICS end-users, ICS end-users' Families
Resulting Change in Well-being	With the commencement of the project activity, the use of the ICS increases fuel efficiency, and an increase in monetary savings is expected. This change is the result of the reduction in the cost of firewood purchases over the lifespan of the project activity due to the increase in fuel efficiency with the project stoves. As per the project activity's Emission Reduction Calculations /06/ for project activities in the Highlands, the quantity of firewood calculated to be saved was 3.81 tonnes/year. Coupled with the Baseline Survey Report where it was found that end-users typically spent K540/year on firewood, the potential savings an ICS end-user could save is K2,116.8 or USD560.84.
VVB Assessment	The validation team validated the information from the document review, sectoral expertise and remote audit interviews with stakeholders and confirms the expected impact of this project on the Affected Stakeholder Group(s). It was observed during remote audit inspections that the beneficiaries observed monetary relief in their usual expenditures /07/. The implementation of project activity is expected to enhance the same across all the localities and provinces where the PA is implemented. The description in section 3.2 of the SD Vista PD /05/ is deemed appropriate and feasible.
Impact #7	Reduction in sales of Firewood for Firewood Vendors
Type of Impact	Negative, actual, direct
Affected Stakeholder Group(s)	Firewood Vendors
Resulting Change in Well-being	In the absence of the project activity, end-users may opt to procure firewood in order to meet the volumetric need for daily cooking and heating practices. With the commencement of the project activity, there is a possibility that there will be a reduction in demand of firewood associated with the improved cookstoves that might affect the vendors selling fuel wood in the project area. However, considering the significant demand-supply gap on fuelwood for cooking needs, decrease in the fuel-wood demand in project location is not likely to have a significant negative impact on wood vendors.

VVB Assessment

The validation team validated the information from the document review, sectoral expertise and remote audit interviews with stakeholders and confirms the expected impact of this project on the Affected Stakeholder Group(s).

The validation team confirms during the remote audit and desk review that a high percentage of the population in PNG purchases or collects wood while using the traditional three stone fire stoves due to high fuel consumption. However, after receiving the project ICS, many households have either reduced purchasing or completely stopped purchasing fuelwood due to comparatively less firewood required for cooking. But since the firewood would still be used for cooking in the ICS, the impact on the wood vendors due to usage of project ICS will not be significantly negative.

The description in section 3.2 of the SD Vista PD /05/ is deemed appropriate and feasible.

3.1.3 Mitigation of Negative Impacts on Stakeholders

The Causal Chain Diagram in Section 2.1.9 of SD VISta PD identifies Negative Impact on livelihood of the local fuelwood vendors as an unintended consequence of the Project Activity. The implementation of the project activity is expected to mitigate this by generating alternative job opportunities for the local population, including women and youth. Likewise, the training imparted to the local population would enhance skill development thus ensuring employability post project activity also.

VVB, on basis of remote audit interviews/11/ and assessment of employee manual/26/, training manual /35/ concludes that PA expects to create jobs for women and youth thus mitigating the unintended negative impact.

Also, considering the significant demand-supply gap and the over reliance of the population of PNG on biomass fuel for cooking needs, decrease in the fuel demand in project location is not likely to have a significant negative impact on fuelwood vendors.

3.1.4 Stakeholder Monitoring Plan

Section 3.3 'stakeholder monitoring plan' of SD VISta PD/05/ describes the identification of the monitored stakeholder groups, the types of measurements, the sampling methods and the frequency of monitoring and reporting. PP will collect data through interviews, surveys, direct observations, group discussions with stakeholders as well as from their financial, health and employment records. PP has visited the intended beneficiaries to carry out identification, sensitization, distribution and further follow up.

The monitoring plan has been summarized and tabulated below for further clarification:

No.	SDG Target	SDG Indicator	Monitoring Frequency	Parameter Monitored	Evidence	Means of Verification
1	1.1	Goal 1: Amount of money saved by families that buy firewood	Annually	Number of cookstoves distributed	Total number of cookstoves distributed. Total number of cookstoves distributed contributes to a reduction of firewood used, which is usually purchased at high prices.	The monitoring approach defined by the PP has been reviewed and found to be appropriate. In the baseline scenario, households either collect wood from the nearby forest or purchase wood, which has been verified through the baseline study/38/ conducted by the PP. Through the CCT conducted by the PP, it is observed that there ICS distributed uses less amount of wood than in the baseline scenario. Hence, there will be reduction in use of fuel and hence, there will be saving in amount of money to purchase the firewood. Hence, amount of cookstove distributed is directly related to the amount of money saved, which is found to be appropriate. The proposed monitoring would support the impacts as defined by the project and for the monitoring of SD Vista Labelled VCUs.

2	3.9	Goal 3: Reduction of fine particulate matter (PM2.5) emissions within households to reduce air pollution within households in the project area.	Annually	Level of fine particulate matter (PM2.5) emissions within households' kitchen area during a cooking activity.	Sampled PM2.5 levels should be below the baseline emission level of 943.3 mg/MJ	The impact will be monitored through during the monitoring survey conducted by the PP during the monitoring period through a handheld air quality sensor, when the ICS were in use by the household. The sampling plan is same as described in VCS PD. The final reading will be the average of all readings taken by the PP. The monitoring approach defined by the PP has been reviewed and found to be appropriate. The proposed monitoring would support the impacts as defined by the project and for the monitoring of SD Vista Labelled VCUs.
3	4.3	Goal 4.1: Number of youth and adults in formal and non-formal education and training in the previous 12 months, by sex	Annually or when there are new hires	Number of participants during training sessions Participation rate is expected to remain constant but may fluctuate slightly depending on the locals available to work.	Signed participation/attendance forms, onboarding documents, employment contracts.	The Impact will be monitored through the number of participants in the training sessions conducted by the PP. The monitoring approach defined by the PP has been reviewed and found to be appropriate. PP will provide the training attendance records, photographic evidence to substantiate the participation in the training sessions and hence, found to be appropriate. The proposed monitoring would support the impacts as defined

						by the project and for the monitoring of SD Vista Labelled VCUs.
	4.4	Goal 4.2: Number of youth and adults with technical and vocational skills, by type of skill within the project area	Annually			The Impact will be monitored through the number of participants in the training sessions conducted by the PP. The monitoring approach defined by the PP has been reviewed and found to be appropriate. PP will provide the training attendance records, photographic evidence to substantiate the participation in the training sessions and hence, found to be appropriate. Additionally, PP has shared the MoU signed with the Western Pacific University (WPU)/55/, which clearly states that Vocational and technical trainings and employment will be provided to the local university students and appointed staff in the area of sustainability, climate change impacts and mitigation strategies and improved cookstoves. The proposed monitoring would support the impacts as defined by the project and for the monitoring of SD Vista Labelled VCUs.

	8.3	Goal 8: Number of people employed, by sector and sex, under the project activity	Annually	Employment contracts	Sampled employment contracts that can be further classified by sector and sex under the project activity.	The impact will be monitored through the employment contracts, onboarding documents. PP aims to generate an estimate of 10 full-time and 120 part-time local employment through the implantation of the project activity. The monitoring approach defined by the PP has been reviewed and found to be appropriate. The proposed monitoring would support the impacts as defined by the project and for the monitoring of SD Vista Labelled VCUs.
4	8.5	8.5.1 Average hourly earnings of employees, by sex, age, occupation, and persons with disabilities	Annually	Employee Payslips, Employment contracts	Sampled Employee Payslips should show fair average earning of employees. Data to be categorized by sex, age, occupation, and persons with disabilities.	The impact will be monitored through the employee payslips and employment contracts. The monitoring approach defined by the PP has been reviewed and found to be appropriate. The proposed monitoring would support the impacts as defined by the project and for the monitoring of SD Vista Labelled VCUs.
5	5.4	5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age, and location.	Annually	Number of trips made to collect firewood per week	Sampled trips should below the baseline scenario of 3-5 times/week. Reduction of trips made per week indicates more time freed	The impact will be monitored through the monitoring surveys conducted by the PP, during which the users were asked questions about the usage of firewood and number of trips

					up to do more productive activities such as part-time work, studying, etc.	conducted to collect the wood. The number of trips will be compared to the baseline scenario 3-5 trips/ week, which has been obtained through the baseline study/38/ conducted by the PP. The monitoring results will be checked at the time of verification to confirm the same. The monitoring approach defined by the PP has been reviewed and found to be appropriate. The proposed monitoring would support the impacts as defined by the project and for the monitoring of SD Vista Labelled VCUs.
6	5.5	Goal 5: Number of women in the project including managerial roles	Annually	Number of women hired for the project activity	Employment contracts	The impact will be monitored through the employment contracts. PP envisages that 50% of the employment will be women. The monitoring approach defined by the PP has been reviewed and found to be appropriate. The proposed monitoring would support the impacts as defined by the project and for the monitoring of SD Vista Labelled VCUs.

The Stakeholder monitoring plan takes into account all stakeholder groups. the impacts expected from the project on them, monitoring indicator, approach, and frequency as well as the procedure for calculating the same. The validation team confirms that the monitoring plan is comprehensive and all subsuming.

3.1.5 Net Positive Stakeholder Wellbeing Impacts

SD VISta PD section 3.4 describes net positive impact on different stakeholders separately.

1. By using efficient combustion of woody fuel, people can avoid high levels of CO, PM 2.5 and other harmful fumes that affect their health.
2. Clean cooking technology can reach more people and satisfy their daily cooking needs in a safe and convenient way.
3. With less fuel wood needed for cooking, families can save money that they would otherwise spend on buying wood fuel.
4. People can free up their time from the tedious and unpaid tasks of collecting wood fuel and cooking and use it for other productive or leisure activities.
5. Training and vocational skills will be provided to local people, which will improve their chances of finding employment and enhance their social and economic wellbeing.

The remote audit observations and interviews along with desk review demonstrates that the project has had a positive impact on stakeholders. The validation team assessed the information and concluded that the justification described in the section is correct and appropriate net impact of project on the stakeholders is positive.

4 BENEFITS FOR THE PLANET

4.1.1 Condition of Natural Capital and Ecosystem Services at Project Start

Section 4.1 of the SD VISta PD/05/ states the condition of natural capital and ecosystem services at the start of the project. It has been validated by the audit team by assessing several reports and scientific writings available online and confirms the appropriateness of these description in the SD VISta PD. The description of the conditions prior to the project starts and the threats faced by the ecosystems were confirmed during the interviews/inspection by validation team through the interviews of end users and the local stakeholders /11/

The validation team concluded that information provided in PD is accurate about conditions at the project start date with respect to natural capital and ecosystem services.

4.1.2 Expected Impacts on Natural Capital and Ecosystem Services

Impact #1	Reduced demand for large diameter non-renewable firewood from live trees due to only needing finger-sized woody biomass, i.e., twigs and crop residues
Type of Impact	Positive, actual, direct
Affected Stakeholder Group(s)	Live plant biomass, Biodiversity and species Richness, Soil and Water Conservation
Resulting Change in Well-being	By slowing the consumption of woody biomass for cooking purposes and by eliminating the need for large diameter woody biomass due to the technology in the ICSs, the resulting drop in demand slows deforestation, thus increasing the above-ground biomass in the adjoining forests near the project area. Reduction in unsustainable harvesting of plant biomass would in turn reduce soil erosion and result in improvement in biodiversity and species richness of the implementation area.
VVB Assessment	This has been verified by the assessment team by interviewing the end users during the remote audit that the firewood consumption has reduced by nearly half since they have started using the project ICS. This has also significantly reduced the time spent in collection of firewood by the households. Thus, the description in section 4.2 of the SD Vista PD /05/ is deemed appropriate.
Impact #2	Avoided emission of GHGs made possible by using ICSs
Type of Impact	Positive, predicted, indirect
Affected Stakeholder Group(s)	Air quality in the households as well as surroundings

Resulting Change in Well-being	The project stoves require less fuelwood for producing the same amount of thermal energy, therefore, resulting in less carbon dioxide emission in the atmosphere. As carbon dioxide is a greenhouse gas, its reduced emission translates to a positive impact on climate change. An annual average of 242,804 tCO₂e is estimated to be avoided due to the inception of the project activity.
VVB Assessment	It is validated through the remote audit interviews and desk review that the project stoves require less firewood to produce same amount of thermal energy therefore resulting in reduction of GHG emissions due to burning of firewood. Each project ICS will lead to saving of ~3.81 tons of wood per stove. The description in section 4.2 of the SD Vista PD /05/ is deemed appropriate.

4.1.3 Mitigation of Negative Impacts on Natural Capital and Ecosystem Services

To assess the measures needed and designed to mitigate any negative impacts on natural capital and ecosystem service validation team assessed the PD, supportive documents, interviewed the stakeholders, including PP, end users, local community leaders and NGOs, and concluded that no negative impacts have been identified on natural capital and ecosystem services on implementation of project activities.

Earthhood assessment team concluded that no negative impacts on natural capital and ecosystem services are ascertained to arise from project activity hence need for mitigation of same wouldn't arise.

4.1.4 Natural Capital and Ecosystem Services Monitoring Plan

The PP has designed a monitoring plan to track the effects of the project activities on the natural capital and ecosystem services, which is described in SD VISta PD section 4.3. The monitoring plan is structured according to the project activities and allows us to track the contributions on the SDG, described in section 1 of the SD-VISta PD, and to support all impacts and claims. All variables are directly linked to the impacts on the planet expected by the project.

All necessary parameters required to support all impacts and claims described in the project's expected and net impacts are contained in the monitoring plan and are clearly described. The desk review and remote audit interview with PP confirms that monitoring arrangements described in the monitoring plan are feasible within the project design and that the PP will be able to implement the monitoring plan.

4.1.5 Net Positive Natural Capital and Ecosystem Services Impacts

It has been validated from Ex-ante ER sheet /06/ that the project generates an estimated 2,992,066 tCO₂e emission reductions over the 10 years project crediting period. A total of 100,000 ICS will be distributed within the country of Papua New Guinea as per the Ex-ante estimate as per the project design.

The year wise estimated GHG Reduction has been given below:

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
------	--

(01/01/2024 to 31/12/2024) Year 1	140,716.61
(01/01/2025 to 31/12/2025) Year 2	266,124.72
(01/01/2026 to 31/12/2026) Year 3	215,631.60
(01/01/2027 to 31/12/2027) Year 4	297,388.88
(01/01/2028 to 31/12/2028) Year 5	281,153.30
(01/01/2029 to 31/12/2029) Year 6	265,784.70
(01/01/2030 to 31/12/2030) Year 7	251,237.50
(01/01/2031 to 31/12/2031) Year 8	237,468.48
(01/01/2032 to 31/12/2032) Year 9	224,436.69
(01/01/2033 to 31/12/2033) Year 10	212,103.29
Total estimated ERs	2,428,045 (Round Down)
Total number of crediting years	10
Average annual ERs	242,804 (Round Down)

No negative ecological impacts have been identified to be arising from the project, during assessment. VVB therefore concludes that the net impact of the project activity on natural capital is positive.

5 VALIDATION CONCLUSION

The Project Participant, Tasman Environmental Markets Asia Pacific, has commissioned the VVB, Earthood Services Private Limited to perform an independent validation of the VCS Project Activity “Improved Cookstove Grouped Project in Papua New Guinea”. This report summarizes the findings of the validation of the project, performed based on SD VISta criteria, as well as criteria given to provide for consistent project operations, monitoring, and reporting.

The validation process was performed based on all guidance and criteria as provided in Sustainable Development Verified Impact Standard /01/ and SD VISta Program Guide /02/2.1.

The review of the Project Description: SD VISta, the project, supporting documents, remote audit, and subsequent follow-up actions (independent research of information) has provided Earthood with sufficient evidence to determine the fulfilment of stated criteria.

Earthood validation approach is based on the understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these. Earthood planned and performed the validation by obtaining evidence and other information and explanations that Earthood considered necessary to give reasonable assurance that reported GHG emission reductions are fairly stated.

The proposed project aims to improve household and community access to improved cookstoves by distributing high thermal efficiency, low greenhouse gas emission cooking stoves known as G3 Burn rocket stove (Improved Cook Stoves) to households of Papua New Guinea. The project is aiming to contribute towards 8 SDGs out of 17 SDGs. This validation was carried out using a risk-based approach. 03 Corrective Action Requests (CAR), 04 Clarification Requests (CL) and 01 Forward Action Request (FAR) were raised and successfully closed during the validation.

VVB confirms that the project complies with the validation criteria for project set out in the Sustainable Development Verified Impact Standard and the SD VISTA Program Guide /02/2.1, including a reasonable level of assurance for the validation.

As a result, the assessment team of Earthood confirms that:

- The project fulfils criteria of Sustainable Development Verified Impact Standard /01/.
- The project is in line with all relevant SD VISTA requirements /01,02/.
- The project SDGs are sufficiently justified in the SD VISTA PD /05/.

The assessment by Earthood also validates the estimated (Ex Ante) value of SDG impacts 1.1, 3.9, 4.3, 4.4, 5.4, 5.5, 7.1, 8.3, 8.5 and 15.2.

Approved by:



Kaviraj Singh

Managing Director

Date: 20-08-2024

APPENDIX 1: COMPETENCE STATEMENT

Competence Statement			
Name	Divij Varshney		
Education	M.Tech. Renewable energy systems B.Tech. Electrical Engineering		
Experience	1.5 years		
Field	Climate Change & Environment / Industry		
Approved Roles			
Team Leader	Yes (VM)		
Validator	Yes (VM)		
Verifier	Yes (VM)		
Local expert	Yes (India)		
Financial Expert	NO		
Technical Reviewer	NO		
TA Expert (X.X)	Yes (VM TA 1.2, 3.1)		
Reviewed by	Shifali Guleria, Quality Manager	Date	16/04/2023
Approved by	Deepika Mahala, Technical Manager	Date	16/04/2023

Competence Statement			
Name	Deepika Mahala		
Education	M. Sc. (Environment Management), GGSIP University B.Sc. Hons. (Chemistry), Sri Venkateshwar College, DU		
Experience	6 Years +		
Field	Climate Change		
Approved Roles			
Team Leader	YES		
Validator	YES		
Verifier	YES		
Local expert	YES (India, Bangladesh)		
Financial Expert	NO		
Technical Reviewer	YES		
TA Expert	YES (TA 1.2 & TA 3.1) (TA 1.1, 13.1, 13.2 – VM only)		
Reviewed by	Shifali Guleria (QM)	Date	28/06/2023
Approved by	Kaviraj Singh (MD)	Date	28/06/2023

Competence Statement	
Name	Vaishali Yadav
Education	M.Sc. environmental studies
Experience	-
Field	Climate Change & Environment / Industry
Approved Roles	

Team Leader	NO		
Validator	NO		
Verifier	NO		
Local expert	NO		
Financial Expert	NO		
Technical Reviewer	NO		
TA Expert (X.X)	NO		
Trainee	YES		
Reviewed by	Shifali Guleria (Quality Manager)	Date	12/07/2023
Approved by	Deepika Mahala (Technical Manager)	Date	12/07/2023

Competence Statement			
Name	Shifali Guleria		
Education	M.Sc. (Environmental Studies and Resource Management), TERI University		
Experience	3+ year		
Field	Climate Change		
Approved Roles			
Team Leader	YES		
Validator	YES		
Verifier	YES		
Methodology Expert	YES (AMS-I.A., AMS-II.G., AMS-II.E., AMS-III.A.V., AMS-I.D, ACM0002)		
Local expert	YES		
Financial Expert	NO		
Technical Reviewer	YES		
TA Expert	YES (1.2, 3.1)		
Reviewed by	Deepika Mahala	Date	18/02/2022
Approved by	Ashok Gautam	Date	18/02/2022

Competence Statement	
Name	Parul Srivastava
Education	PhD Forest Ecology and Environment M.Sc. Botany B.Sc. Botany and Chemistry
Experience	20 years
Field	Forestry
Approved Roles	
Team Leader	NO
Validator	NO
Verifier	NO
Methodology Expert	NO
Local expert	NO
Financial Expert	NO
Technical Reviewer	NO

TA Expert (14.1)	YES		
Other roles	Yes (SDVISTA SS 1)		
Reviewed by	Shifali Guleria (Quality Manager)	Date	13/04/2023
Approved by	Deepika Mahala (Technical Manager)	Date	13/04/2023

Competence Statement			
Name	Sumit Kaushik		
Country	India		
Education	Ph.D. (Social Entrepreneurship) research (Ongoing) Ph.D. (Business Administration) MBA (Communication Management) BA (Journalism and Mass Communication)		
Experience	10 Years +		
Field	Public Policy and Communication		
Approved Roles			
Team Leader	NO		
Validator	NO		
Verifier	NO		
Methodology Expert	NO		
Local expert	NO		
Financial Expert	NO		
Technical Reviewer	NO		
TA Expert	NO		
Other roles	YES (SDVISTA 10)		
Reviewed by	Shifali Guleria (QM)	Date	01/08/2023
Approved by	Deepika Mahala (TM)	Date	01/08/2023

Competence Statement	
Name	Ranjan Singh
Education	BSc (Physics), MBA (Marketing)
Experience	13 Years
Field	Power, Utilities and Renewables
Approved Roles	
Team Leader	YES (VM only)
Validator	YES
Verifier	YES
Local expert	YES (India)
Financial Expert	NO
Technical Reviewer	NO
TA Expert (X.X)	YES (TA 1.2)

Other roles	Yes (SDVISTa SS 2, 4)		
Reviewed by	Shifali Guleria (Quality Manager)	Date	29/06/2023
Approved by	Deepika Mahala (Technical Manager)	Date	29/06/2023
Competence Statement			
Name	Avinash Kumar		
Country	India		
Education	MBA (Marketing) M.Sc. (Environment)		
Experience	13 Years +		
Field	Sustainability & Climate Change		
Approved Roles			
Team Leader	NO		
Validator	NO		
Verifier	NO		
Methodology Expert	NO		
Local expert	NO		
Financial Expert	NO		
Technical Reviewer	NO		
TA Expert	NO		
Other roles	YES (SDVISTa SS 3, 5, 6, 7, 8, 9, 12)		
Reviewed by	Shifali Guleria (QM)	Date	10/08/2023
Approved by	Deepika Mahala (TM)	Date	10/08/2023

APPENDIX II: ABBREVIATIONS

Abbreviations	Full texts
AFOLU	Agriculture, Forestry and Other Land Use
BE	Baseline Emission
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CL	Clarification Action
CO ₂	Carbon dioxide
CO _{2e}	Carbon dioxide equivalent
DOE	Designated Operational Entity
DVR	Draft Validation Report
EB	CDM Executive Board
EF	Emission Factor
EIA	Environmental Impact Assessment
ER	Emission Reduction
Earthood	Earthood Services Private Limited

FAO	Food and Agriculture Organization
FAR	Forward Action Request
GHG	Greenhouse gas(es)
GP	Grouped Project
ICS	Improved Cook Stoves
ILO	International Labour Organization
IPCCC	Intergovernmental Panel on Climate Change
MP	Monitoring Plan
MR	Monitoring Report
NA	Not Applicable
PA	Project Activity
PD	Project Description
PE	Project Emission
PP	Project Participant
QA/QC	Quality Assurance / Quality Control
QMS	Quality Management System
SD VISTA	Sustainable Development Verified Impact Standard
SDG	Sustainable Development Goals
TR	Technical Review
TSF	Three-stone fire stove
UNFCCC	United Nations Framework Convention on Climate Change
VCS	Verified Carbon Standard
VCS-PD	VCS – Project Description
VCU	Verified Carbon Unit
VVB	Validation/verification Body
XLS	Emission Reduction Calculation Spread Sheet

APPENDIX III: REFERENCES

S.No.	Title of Document	Version	Provided By
1.	Sustainable Development Verified Impact Standard (dated 22 January 2019)	Ver 1.0	VCS
2.	2.1 SD VISTA Programme Guide (dated 22 January 2019) 2.2 SD VISTA Programme Definition (dated 22 January 2019)	Ver 1.0 Ver 1.0	VCS
3.	VCS Standard	Ver 4.4	VCS
4.	SD VISTA Project-Description-Template	Ver 1.0	VCS
5.	SD VISTA PD	Ver 1.2, dated 24/07/2024	PP
6.	Ex Ante ER Sheet- VCS	-	PP

7.	Remote Audit Records	-	VVB
8.	VMR0006 Methodology	1.2	VCS
9.	Earthhood contract with PP	-	VVB
10.	Anti-bribery and corruption Policy	-	PP
11.	Interview Records	-	VVB
12.	CDM Guidelines Sampling and surveys of CDM project activities and programmes of activities Sampling and surveys for CDM project activities and programmes of activities	Ver 4.0 Ver 9.0	CDM
13.	Project Registry Webpage (Last accessed: 26/12/2023) https://registry.terra.org/app/projectDetail/VCS/4416	-	VCS
14.	TEM's Modern Slavery Policy	-	PP
15.	Stakeholder Consultation Report	Oct 2022	PP
16.	<i>Baseline assessment report</i> (Baseline Survey database)	Oct-Nov 2022	PP
17.	Declaration/ confirmation by PP that the project activity will not seek credits under any other program	-	-
18.	Declaration confirming that PP will own the Verified emission reductions	-	-
19.	Declaration that the stoves are provided at zero cost	-	05/05/2023
20.	Earthhood's Remote Audit	-	22/09/2023
21.	WBT test report by CREEC Laboratory	-	25/05/2022
22.	VCS Final Validation Report	-	-
23.	LSC Feedback forms	-	-
24.	Email invitations, Newspaper advertisement and posters for LSC	-	-
25.	Attendance of LSC attendees, LSC Report, photos and material	-	-

26.	TEM's Employee Handbook	-	-
27.	TEM's Website announcement	-	-
28.	Standard of sampling and survey of CDM Project activities and programme of activities	Version 09.0	-
29.	The Guidelines for sampling and survey for CDM Project activities and programme of activities	Version 04.0	-
30.	Pilot Stove distribution database	-	-
31.	Grievance Redress Policy and Procedure Manual,	Version 1.2	-
32.	Training Records/pictures	-	-
33.	TEM-APAC Security Review	SecRisk	17/05/2023
34.	SPFEC Peer Review of fNRB calculations for the highlands region and PNG		17/03/2023
35.	End-user Training Manual	-	-
36.	Pilot Study conducted by TEM	-	-
37.	TEM letter of support	-	-
38.	Cookstoves Highlands region Baseline survey report	v2.0	March 2023
39.	TEM Simi Oa contract	-	-
40.	Verra's Letter of Exemption for the on-site visit	-	30/08/2023
41.	Old UN Default values (EB67, Annex 22)	-	-
42.	Hanson, L., Allen, B., Bourke, M., & McCarthy, T. (2001). Papua New Guinea Rural Development Handbook. (1st ed.) Australian National University.	-	2001
43.	Konedobu, P.N.G.: Energy Planning Unit, Department of Minerals and Energy, 1982		
44.	ACIAR (Australian Centre for International Research). 2013. Promoting diverse firewood production systems in Papua New Guinea	-	2013
45.	PNG labour legislation		-
46.	IPCC Guidelines	-	-

47.	Confirmation Letter by PNG Forest Authority	-	2019
48.	Page et al., (2016)	-	2016
49.	Murphy (2009) What is free about fuelwood? A critique of the value of fuelwood in the rural and squatter settlement households in the Eastern Highlands of Papua New Guinea	-	2009
50.	FAO (2020). Global Forest Resource Assessment, (FRA) PNG Report	-	2020
51.	Manufacturer's Specifications	-	-
52.	2011 National Population and Housing Census. Papua New Guinea 2011 National Report	-	-
53.	IRENA (International Renewable Energy Agency), 2013. Report	-	-
54.	End user beneficiary sample agreement. Signed end-user agreement for the cookstoves distributed during the pilot phase	-	-
55.	TEM's MoU with Western Pacific University (WPU)	-	-
56.	Clean Cooking Catalog http://catalog.cleancookstoves.org/stove_details.html?stove_id=stove_A8WGMJX	-	-
57.	https://data.unicef.org/how-many/how-many-children-live-in-poverty/	-	-
58.	https://www.mypsup.org/countries/Papua_New_Guinea#:~:text=Some%2080%20per%20cent%20of,countr y%20has%20several%20urban%20centres.	-	-
59.	World Health Organization (WHO), Health and environment scorecard: Papua New Guinea, Air Pollution	-	-
60.	Anker Living Income Reference Value: Rural Papua New Guinea (2020) (No.20-03-20). Universidad Privada Boliviana. pp10	-	2020
61.	WWF: Papua New Guinea, Area of Forests	-	-
62.	Invitation letter to NGO 'Mt Giluwe Conservation Foundation'	-	12/02/2023

	Confirmation letter from NGO 'Mt Giluwe Conservation Foundation'		10/03/2023
63.	Letter of Support from NGO 'Mt Giluwe Conservation Foundation'		04/03/2023
64.	Approvals from the local authorities: 1. Southern Highlands Province, Approval of Pilot Study (File name: Approval Pilot study SHP.pdf) 2. Western Highlands Province, Approval of Pilot Study (File name: Approval Pilot study WHP.pdf)		23/09/2022 27/09/2022

APPENDIX IV: FINDINGS

CAR: Corrective Action Request

CL: Clarification Request

FAR: Forward Action Request

Table 1. Remaining FAR from validation and/or previous verification

FAR ID		Section no.		Date : DD/MM/YYYY
Description of FAR				
NA				
Project participant response				Date : DD/MM/YYYY
NA				
Documentation provided by project participant				
NA				
VVB assessment				Date: DD/MM/YYYY
NA				

Table 2. CAR from this verification

CAR ID	01	Section no.	2.1	Date : 22/11/2023
Description of CAR				
Apart from the 'Causal Chain' diagram provided in Appendix A, the PP shall include under section 2.1.9, a detailed description and list of all direct positive and negative, intended, and unintended consequences of project activities and impacts related to People and their Prosperity and Planet.				
Project participant response				Date : 28/11/2023
The Sustainable Development Outputs table has been added under Section 2.1.9.				
Documentation provided by project participant				
- Updated SD VISTA PDD				
VVB assessment				Date: 05/12/2023
The PP has revised the PD to reflect the description and list of all consequences resulting from the proposed project activity. The assessment team found the 'Causal Chain' section to be adequately addressed in the document. The description added has been reviewed and found to be appropriate and inline with the causal chain in the appendix 1 of the SDVista PD.				
Thus, CAR#01 is closed.				

CAR ID	02	Section no.	3.2	Date : 22/11/2023
Description of CAR				
In Section 3.2, the PP is requested to add the table regarding the negative impact on the firewood vendors, for whom the demand will be reduced.				
Project participant response				Date: 24/11/2023
Table has been included into the section regarding the negative impact on the firewood vendors.				
Documentation provided by project participant				
- Updated SD Vista PDD				
VVB assessment				Date: 05/12/2023
PP has added the relevant table to account for the negative impact resulting from the proposed project activity. The description of the impact has been reviewed and is found to be appropriate. The revision is found to be consistent with the template requirements.				
Thus, CAR#02 stands closed.				

CAR ID	03	Section no.	2.1.8, 4.2, 3.2	Date: 22/11/2023
Description of CAR				
The following information is found missing in the PD:				
1. Section 2.1.8 of PDD shall include brief summary of the Social, Economic, and natural capita conditions at the start of the Project activity as required by the template guideline. PP shall indicate the aforementioned three conditions explicitly. 2. Expected Impacts on Stakeholder/Natural Capital and Ecosystem Services: Section 3.2 and 4.2 of project description does not include information regarding magnitude of 'Resulting Change in Well-Being' as per required by the template guidelines. The Resulting Change in Well-being shall describe characteristics and magnitude of impact on each stakeholder group.				
Project participant response				Date: 24/11/2023
1. Brief summary of Social, Economic, and natural capita conditions at the start of the Project activity has been included into the section. 2. Quantifiable data and estimations have been added into the sections as much as possible.				
Documentation provided by project participant				
- Updated SD Vista PDD				
VVB assessment				Date: 05/12/2023
1. The PP has summarized (with appropriate sub-heads) the social, economic, and natural capita conditions that existed at the start of the project. 2. The tables in both the sections (3.2 and 4.2) have been revised to account for the characteristics and magnitude of the Resulting Change in Well-being of all the stakeholders.				
The assessment team found the changes to be in line with the template guidelines.				
Therefore, CAR#03 stands closed.				

Table 3. CL from this verification

CL ID	01	Section no.	1	Date : 22/11/2023
Description of CL				

The following changes should be made to the table in section 1 of the PD:

1. According to section 1 of latest applicable template v1.0 of the SD VISta Project Description, an estimate of the project's quantifiable contributions to the specific targets of the Sustainable Development Goals (SDGs) over the project lifetime is to be provided. The PP is requested to reformulate and make the SDG contributions included in rows 2, 4, 5, 6, 7, 9, 10 and 11 of section 1 quantifiable.
2. Improved cookstoves are not on the list of services provided in the SDG metadata, therefore a self-defined indicator for number of cookstoves should be used in Row 1 of the table provided in section 1.

PP shall review and update accordingly.

Project participant response	Date : 24/11/2023
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1. Row 2,4,5,6,7,9 and 10 have been reformulated to estimate the project's quantifiable contribution to the specific targets. Row 11 has been removed as it does not meet the UN Metadata's specifications. The numbering of the rest of the row has been updated accordingly.
2. Self-defined indicator has been included for the number of cookstoves distributed.

Documentation provided by project participant
--

- Updated SD Vista PDD

VVB assessment	Date: 05/12/2023
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1. PP has updated the table to provide the estimated project contribution by the end of project lifetime. However, the quantifiable estimates for Row 2 and 6 has not been defined properly. PP is requested to take corrective action. OPEN
2. The PP has revised the PD to reflect a Self-defined indicator in section 1. The revision has been found to be correct. CLOSED

Therefore, CL#01 stands OPEN.

Project participant response	Date : 08/12/2023
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1. Rows 2 and 6 have been updated to provide more clarity on quantifiable estimates.

Documentation provided by project participant
--

- Updated PDD

VVB assessment	Date: 14/12/2023
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The PP has appropriately clarified upon the estimates for SDG Target 2.1 and 5.4. No further action is required from the PP's end and therefore, the finding CL#01 stands closed.

CL ID	02	Section no.	2	Date : 22/11/2023
Description of CL				
In section 2.1.11, The PP is requested to elaborate and specify the pre-established sampling protocol that will be followed as a part of Routine Spot Audit Evidence and follow ups during each monitoring period to monitor the SDG parameters.7				
Project participant response				Date : 27/11/2023
Routine spot evidence section has been added under section 2.1.11, including table of parameters that should be monitored for co-benefits.				
Documentation provided by project participant				
- Updated SD VISta PDD				
VVB assessment				Date: 05/12/2023
The PD has been revised to add the relevant details related to the Routine Spot Audit Evidence required for monitoring of the present project activity. Further, the table of parameters has been checked and found to be appropriate.				
Therefore, CL#02 stands closed.				

CL ID	03	Section no.	1	Date : 22/11/2023
Description of CL				

Project Description Document does not provide clarification regarding direct impact of Project Activity on SDG 4.4.1 and 8.3.1. PP shall justify and provide supportive of how it will assist in improving employability, given PD mentions no information of baseline unemployment.	
Project participant response	Date : 24/11/2023
Additional information has been included in Rows 5 and 9 of Section 1.	
Documentation provided by project participant	
- Updated SD Vista PDD	
VVB assessment	Date: 05/12/2023
The PP has provided the details for employment generation as a result of the present project activity. It is reflected in Rows 5 and 9 of Section 1 of the project description. Therefore, the finding CL#03 stands closed.	

CL ID	04	Section no.	RSV	Date : 05/12/2023
Description of CL				
During the remote audit of the Western Highlands Provinces (WHP), it was observed that a few recipients (under the pilot-run) owned and used gas stoves and electric stoves, in addition to the ICS. The PP is requested to clarify upon the claimed identified baseline and the recipient selection criteria for the distribution of ICS under the current project.				
Project participant response				Date : 08/12/2023
A third-party consultant team contracted by TEM conducted a baseline survey during October 2022 with the results included in a report that was audited by an esteem in country forester (also included in the evidence provided). File Reference: <u>Cookstoves Highlands region report March 2023.pdf</u> . As established by VCS the purpose of the baseline is to gather information on the current cooking practices in these local communities to understand their cooking behaviours and practices. This, in turn, will enable a better understanding of the additional of the proposed project activity.				
The results from the baseline survey showed that 98% of the houses involved in the study used firewood as a primary source of energy, whereas only 2% of total houses located in semi-urban settlements mentioned the use of electricity or gas as secondary energy sources (p. 25).				
The inclusion of these houses in our baseline may have slightly reduced our B _{old} estimation on firewood consumption/house/year. However, we have decided to take a conservative approach without foreseen any changes in the proposed estimates.				
TEM will apply a "house selection criteria" during the selection of houses. This process will serve as confirmation that each house is aligned with the baseline scenario set out for the project activity.				
To identify the houses at the point of the ICS distribution, we will use a questionnaire including binomial questions (yes/no answers) to access the house's primary method of cooking fuel as follows:				
1. Do you cook any meals in the house without using firewood? 2. Do you use charcoal for cooking in your house? 3. Do you use electric cooking in your house? 4. Do you use LPG for cooking in your house?				
If any of the responses to these questions indicate that firewood is not the main fuel used in the house for cooking, they will not be selected to participate in the proposed project activity.				
Documentation provided by project participant				
- Findings List				
VVB assessment				Date: 14/12/2023
The information provided by the PP has been reviewed and the selection criteria that will be implemented is found to be appropriate.				
Hence, CL#04 stands closed.				

Table 4. FAR from this verification

FAR ID	01	Section No.	NA	Date : 08/12/2023
Description of FAR				
In Line with Section 3.5.6 of the SDVista Program Guide v1.0, during the first verification of the project activity, the next verifying VVB shall conduct an on-site visit of the project activity to reassess the following information provided by the project proponent: Project Location, Baseline Scenario, Stakeholder Engagement and consultation, Conditions of stakeholders at project start, conditions of natural capital and ecosystem services at project start.				
Project participant response				Date : 08/12/2023
<i>Project Participant notes this FAR for an on-site visit to be conducted during the first verification of the project.</i>				
Documentation provided by project participant				
VVB assessment				Date: DD/MM/YYYY

e.g., there is no FAR from this validation.